IMPLEMENTATION OF A QUALITY DATA MONITORING SYSTEM FOR INCOMING QUALITY CONTOL (IQC) IN MANUFACTURING

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IMPLEMENTATION OF A QUALITY DATA MONITORING SYSTEM FOR INCOMING QUALITY CONTROL (IQC) IN MANUFACTURING

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Data Science Project Report submitted in fulfilment of the requirements
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ABSTRAK

Dalam industri pembuatan, integrasi teknologi terkini adalah penting untuk meningkatkan kecekapan pengeluaran dan mengekalkan piawaian kualiti yang tinggi. Namun begitu, syarikat seperti TT Electronics menghadapi cabaran dalam Kawalan Kualiti Kemasukan (IQC) kerana kekurangan saluran data bersepadu dan masih bergantung kepada proses manual yang tidak stabil, memakan masa dan mudah terdedah kepada kesilapan. Objektif kajian ini adalah untuk melaksanakan pangkalan data automatik, membangunkan keupayaan ramalan, dan mewujudkan papan pemuka interaktif bagi sistem pemantauan data kualiti IQC yang menyeluruh. Kaedah yang digunakan melibatkan prapemprosesan data yang kukuh serta penggabungan pelbagai jenis data seperti metrik pengeluaran, data pemeriksaan berbentuk teks dan visual. Bagi menganalisis data pengeluaran dan mengenal pasti ketidakkonsistenan, kaedah kecerdasan buatan (AI) lanjutan termasuk pembelajaran mesin (ML) digunakan. Sistem ini dibangunkan dalam persekitaran pengeluaran sebenar, dengan maklum balas dikumpul secara berterusan untuk menambah baik keberkesanan. Kajian ini membantu pengilang mencapai pembuatan berasaskan data dan membuat keputusan secara masa nyata dengan menghapuskan ketidakcekapan dalam proses IQC. Pendekatan yang dicadangkan mengurangkan kebergantungan terhadap proses manual untuk meningkatkan kualiti pengeluaran secara keseluruhan dan memudahkan peralihan ke arah Industri 4.0.

ABSTRACT

In the manufacturing industry, the integration of cutting-edge technology is essential for enhancing production efficiency and maintain high-quality standards. However, companies like TT Electronics face challenges in Incoming Quality Control (IQC) because they lack an integrated data pipeline and reliance on unstable manual processes which are time-consuming and error. The objective is to implement an automated database, develop prediction capabilities, and create an interactive dashboard for a comprehensive IQC quality data monitoring system. The methodology involves combining data and preprocessing various of data such as robust data preprocessing, combining production metrics with textual and visual inspection data. To analyse production data and identify inconsistencies, advance artificial intelligence (AI) method including machine learning (ML) are used. This system is developed in a real-world production environment, with feedback collected to improve effectiveness over time. This study helps manufacturers achieve real time and data driven decision making by eliminating inefficiency in IQC processes. The suggested approach reduces reliance on manual process to enhance overall production quality and the proposed system facilitates the transition to Industry 4.0.

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CHAPTER 1

INTRODUCTION

1.1 Research Background

The centre of Industry 4.0 also known as the fourth industrial revolution, technological advancements in data analytics, collaboration, scalability, and networking promotes innovation and change our perception of how goods and services are manufactured. Industry 4.0, smart manufacturing, and industrial internet-of-things (IIoT) describe how information and operation technology come together to monitor physical manufacturing process and use data to make predictive decision that lower operation cost (Tay et al., 2021). Maintaining quality during manufacturing is crucial, and Quality 4.0 technology can help overcome significant quality-related challenges. Quality 4.0 is a rational strategy to solve a range of technological issues as well as strategic and cultural activities (Javaid et al., 2021). Predictive quality analysis is a method used by producers to forecast the quality of resources, parts and good that are already available in manufacturing processes.

Quality is one of the competitive advantages of products. Enterprise's probability can be increased by keeping proper balance between production efficiency and quality. A key component for manufacturing is quality assurance (QA), which makes sure that goods constantly fulfil established standards of quality before they are delivered to customers. QA involves developing systematic procedures to prevent faults throughout manufacturing, in contrast to quality control (QC), that has emphasis on defect detection. This proactive strategy aims to maintain a high degree of product reliability and customer satisfaction by developing clear processes, documentation standards, and continuous improvement practices. Adding procedures like Incoming Quality Control (IQC), In-Process Quality Control (IPQC), and Final Quality Control (FQC) to IQC and QA results in a staged approach to quality assurance. IQC stands for the quality assurance and verification of components, products, or raw materials that have been purchased. IQC is a method of quality control that involves sampling the raw materials that suppliers send. Monitoring the production process is necessary for IPQC in order to spot problems early on and address them before they affect the next steps. Before a product is shipped, FQC is the last check to make sure it meets quality standards.

Therefore quality assurance techniques in smart manufacturing in the industrial sector are fundamental to the research, it is important to understand how automation technologies impact industry control processes. Modern technologies are approaches for field application which examined production processes by identify Artificial Intelligent (AI) (Galindo-Salcedo et al., 2022). Hence, the goal of this project is developing an automated quality detection system that can recognize and display patterns in production data TT Electronics. This research has been greatly influencing by the company dedication to utilizing Industry 4.0 technology, which offer a practical setting for the implementation of automated quality detection systems.

The data preprocessing step aims at attempt to provide a list of detected signals (features) characterized by several of information a single study can identify thousands to tens of thousands of features, depending on the samples examined. According to Lennon et al. (2024), they can then be grouped and aligned across batches and analyses. The quality of feature detection is crucial for feature integration in the data preprocessing step. Overall, the preprocessing step is the primary disadvantages are the possibility of incomplete peak-picking and features extraction, particularly for detection of low abundant compounds.

Many manufacturing facilities, including TT Electronics, have developed data warehouses, which act as central databases for organizing and storing huge amounts of manufacturing data, to further increase efficiency. Manufacturers may monitor quality indicators continuously and react to new problems more quickly with the help of a well-designed data pipeline that makes use of a data warehouse. Better data management is made possible by this method, which also shortens the time it takes for data to become available and gives quality control procedures real-time insight. It is commonly known that data warehouses is decision support tools for analyzing large amounts of alphanumeric data that are represented using a multidimensional model that establishes the ideas of dimensions and facts (Berrahou et al., 2015). It is true that analyzing the quality of data in a multi-source setting requires systematic thought, particularly in decision-support systems.

1.2 Problem Statement

In manufacturing industry, sustaining competitive advantage on maintaining constant product quality. Incoming Quality Control (IQC) in TT Electronics is facing significant challenges the lack of efficient data integration throughout its quality control procedures. Lack of a consistent data pipeline results in a lot of manual labour, inconsistent data, and ineffective reporting, all of which make it more difficult to identify quality problems in a timely manner. Raw IQC data sometimes contains information that makes it difficult to use directly for analytics without requiring time-consuming treatment. These delays and potential mistakes in quality control affect both customer happiness and operational efficiency.

Furthermore, the current static reporting system do not support continuous quality assessment because they are unable to use integrated and historical data for proactive decision making and trend analysis. Quality abnormalities may therefore remain hidden until much later, which could affect the overall efficacy of the process and the dependability of the final product. For IQC to provide effective, precise quality control, a simplified data integration pipeline with strong preprocessing and real-time quality monitoring capabilities is necessary. This system would eventually raise failure rates, improve process effectiveness and develop more customer trust by proactively identifying and resolving quality issues.

1.3 Research Questions

1. How can an automated quality data monitoring system be developed to effectively track Incoming Quality Control (IQC) inspections at TT Electronics?

2. How can an ETL pipeline be effectively implemented to extract, transform, and load IQC inspection data into a centralized database?

3. What is the predictive performance of PPM defect rates based on past IQC inspection records?

4. How can a real-time reporting dashboard be designed to visualize inspection results, defect trends, and predictive outcomes for better decision-making?

1.4 Research Objectives

1. To develop an automated quality data monitoring system for tracking Incoming Quality Control (IQC) inspections using ETL pipeline.

2. To implement an ETL pipeline that extracts, transforms, and loads IQC inspection data into a centralized database for structured reporting.

3. To build a predictive model that predict the PPM defect rate of incoming material lots based on historical inspection data using Random Forest in Machine Learning.

4. To design a real-time reporting dashboard that visualize inspection trends, defect rates, and predictive insights using Power BI.

1.5 Research Scopes

This study focuses on incoming quality control (IQC) process in TT Electronics. The primary scope of this study to implement a system that automates the reporting of IQC inspections, amin to enhance the tracking the defect rates and inspections outcome for all incoming materials in the first quarter of 2025, with data collection continuing throughout the year. The system will be designed to operate with a defined frequency of inspection reporting, daily report, weekly trend analysis and monthly summary report.

The data collected from various source such as IQC inspection log, defect report and supplier record. An ETL (Extract, Transform, Load) Pipeline will be implemented to systematically gather, clean and store this data into a centralized databased for consistency and structured analysis.

A Random Forest model will be applied to predict quality outcomes and identify key factors associated with inspection failures. The performance of this model will be evaluated using standard metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).

The analysis will be conducted using tool Excel which is Visual Basic for Applications (VBA) and Power BI, which will enable the visualization of trends and patterns in the dataset. This tool will facilitate interactive dashboards for real-time insights and reporting. Scope includes defining parameters for pattern recognition and developing visualizations to support proactive quality assurance.

1.6 Significance of study

This study has valuable significance for quality management operations at TT Electronics, especially incoming quality control. The main contributions of this study are the automation of data collection and reporting, the system proposed guarantees minimal interference of human effort and time on data handling tasks. This speeds up inspection advances, cuts down human errors, and allows quality employees to engage in more important matters such as analysis and process improvement.

These interactive dashboards and automatic reports allow the system to store and display an inspection's findings, defect trends, and supplier performances instantly. Quality engineers and decision-makers can, therefore, react immediately to quality issues and enforce remedial measures without delay.

The selection of a centralized database and structured ETL pipeline guarantees uniformity for data formatting, reporting, and information across departments and this, in turn, makes inspection data reliable and facilitates internal communication and audit processes.

Implementation of predictive analytics with the kind of Random Forest-based machine learning models allows the organizations to become data-oriented in their decision-making. It is possible to mine defect anticipation and key factors that influence defect occurrence, thereby benefiting quality control interventions and long-term process improvements.

1.7 Thesis Organization

This dissertation is organized into five chapters to provide a systematic overview of the research. Chapter 1 presents the study by summarizing the research context, pinpointing the main issue, crafting the research questions and aims, and clarifying the study's scope and importance. Chapter 2 centres on data gathering and acquisition, detailing the origins of IQC data, the extraction of pertinent variables, and the application of automation tools like VBA for streamlined data integration. Chapter 3 outlines the research methods, explaining the execution of the ETL pipeline, data preprocessing strategies, and the use of the Random Forest algorithm to forecast PPM defect rates. Chapter 4 focuses on the outcomes from the created system, assesses the efficacy of the predictive model, and presents real-time data via interactive Power BI dashboards. In conclusion, Chapter 5 wraps up the research by highlighting the main findings, discussing the study’s limitations, and offering suggestions for future enhancements to the system data monitoring system to improve IQC processes at TT Electronics.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter review of existing literature to understand the current trends, practices, and challenges in quality assurance and control within the context of Industry 4.0 and smart manufacturing. It explores how advancements in data analytics and automation contribute to quality detection systems while identifying gaps in research relevant to the development of optimized solutions for TT Electronics.

2.2 Quality Control and Quality Assurance (QA) in Manufacturing

Quality Control (QC) and Quality Assurance (QA) are process use in many industries with other standard, operational efficiency, customer satisfaction and consistent quality. Process control and use of methods, tools, and best practices can then be adopted to monitor and assess evaluation in processes, product, service. Implementing QA and QC can minimize the defect and waste and increase customer satisfaction and competitive edge. QA and QC methods are important in production in ensuring reliability and quality of products. The method includes quality audits, product testing, process control, and raw material inspection. Manufacturing companies are able to identify detect flaws, lower variability, streamline production procedures, and enhance overall quality by implementing statistical process control, failure mode and effect analysis, and lean manufacturing concepts.

QC in manufacturing is to make sure that the products meet specific requirements and specifications along with safety standards and expectations. QC is generally systematic activities aimed at monitoring and maintaining product quality throughout the production process. These activities typically include inspections, testing, process validation, and the use of tools and software to detect and correct issues before products reach the market (Lennon et al., 2024).

In modern manufacturing, QC has evolved with the integration of cutting-edge technologies such as machine learning, automation, and real-time data analysis. One is predictive analytics, which would use historical data to provide insight into the likelihood of future quality problems so that manufacturers can detect defects and inefficiencies earlier in the production cycle. Automated systems, like quality detection system, collects real-time data from production lines to identify patterns of abnormal behavior and alert operators to potential faults, thus avoiding extensive about potential defects, reducing the need for manual intervention.

The good model to implement QC system in manufacturing is for defect minimization purpose, reducing waste, ensuring compliance with regulations, and maintaining customer satisfaction. The future of quality control in manufacturing is set to become even more automated, data-driven, and efficient, rapid technology advances.

Table 2. 1 Findings of Quality Control and Quality Assurance

No	Tittle	Author	Finding	Gap
1.	Quality Management, Quality Assurance, and Quality Control in Medical Physics	Amurao et al. (2023)	Clarifies definitions of QM, QA, and QC in medical physics; provides examples of their implementation in ACR standards to enhance safety and quality.	Limited exploration of how these frameworks adapt to evolving technologies and diverse global healthcare settings.
2.	Quality Control and Quality Assurance	T. Ahmad (2024)	Relationship with quality management is TQM, SPC, Lean Six Sigma and continuous improvement.	Lack of application examples or case studies in specific industries to validate the approaches.
3.	Improving the defect detection performance of Incoming Quality Control	de Vent et al.(2024)	The study developed a predictive model using various machine learning techniques, including Logistic Regression, Decision Tree, SVM, XGBoost, and CatBoost.	limitations in their predictive power, accuracy, scope, or applicability to certain scenarios.

According to ISO 9000, QA is a component of quality management that focuses on making sure quality standards are fulfilled. Quality assurance includes all organized, systematic, and useful efforts aimed at demonstrating the adequacy of the quality of a process's output (Amurao et al., 2023). The process has been constructed up of connected resources and actions that transform inputs into outputs, according to the International Electrotechnical Commission. By considering elements including tools, processes, and systems, QA assesses the accuracy of these outputs.

Shewhart was the first to suggest statistical process control (SPC), which is now widely acknowledged as an efficient quality control method (Wang et al., 2024). SPC investigates the connections between statistical features and targets based on statistics and offers a warning system that quickly monitors the process. Due to research and practice, several advanced monitoring charts have emerged in recent decades, such as a cumulative sum, exponentially weighted moving average, and synthetic control charts (Wang et al., 2024). These methods have made it easier to apply engineering process controls (EPCs) and allowed SPC systems to enable noticeably reduce defects per million opportunities.

Incoming Quality Control (IQC) is a fundamental process within Electronics Manufacturing Services (EMS), this is the first step for ensuring product quality by evaluating purchased components and materials before they are introduced into production. IQC play a vital role in safeguarding manufacturing operation from defective inputs, particularly in manufacturing environment where precision, reliability and compliance with customer specifications are essential.

IQC operations begin when good are received in the warehouse. Upon arrival, all incoming materials are logged in the warehouse management, which is synchronizes in near real time with the system (de Vent et al., n.d.). It is combination of automated selection method and manual intervention is used to determine which items are subject to inspection.

Once selected for inspection, items temporarily stored in a specified warehouse location. The IQC team, which made up of committed inspectors and support personnel, referred to inspection instruction rated by quality engineers using specialized software like Compact (de Vent et al., n.d.).Inspection may include visual assessment, dimensional verification and other functional test depending on the component category. The IQC staff chooses the sampling size based on the item type and past defect data, and on average, three purchasing lines are examined every day. If a defect is found, a quality engineer reviews the complaint and decides whether it is valid and what to do next. The complaint is then recorded in the report.

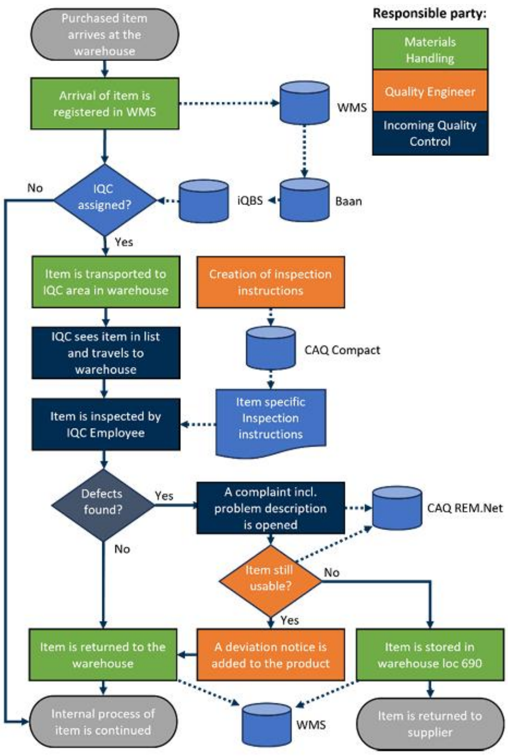

Figure 2. 1 Shows a flow chart of the incoming quality control and item rejection process. Source: (de Vent et al., n.d.)

2.3 Data Warehouses and Pipeline Design for Manufacturing Data

The Data Warehouse system has two main architecture the first is data flow architecture and device architecture is the second (Merceedi et al., 2022). The first architecture describes how to store data is structured and organized in the data warehouse. It represents the course by which the data flow from the source system to the end users. The second architecture describe how the serves, network, application, storage and client are organized or clients are physically structured. To effective implement architectures, it essential to consider the design methodologies which is Inmon and Kimball methodologies, which will be discussed further in Section 2.3.1.

The creation of a Quality-Aware Spatial Data Warehouse (QASDW) to solve difficulties in evaluating environmental data (Berrahou et al., 2015). Robust multidimensional analysis is made possible by the QASDW's direct integration of geographical, time-based, and thematic quality factors into its design. This method use data quality indicators as filters to improve decision-making, in contrast to conventional ETL procedures that concentrate on data cleaning. The processes involved in integration data in Extract, Transform and Load (ETL) and Extract, Load, transform (ELT) methodologies, which will be explored in Section 2.3.2. The warehouse utilizes use of PostgreSQL and Mondrian server to provide querying based on quality parameters, including measurement timing and spatial proximity. Experts can spot important patterns and areas for intervention with the use of complex visualization tools, such as color-coded cells, which improves resource management and inspection efficiency.

Data pipelines would be deployed as software services that operate without any problem in the Cloud or inside on-premises servers (Poojara et al., 2022). However, a lot of IoT applications are event-driven and require real time operation is necessitates to build expensive and massive data processing clusters (like Apache Spark, Flink, and Storm) to handle huge amounts of IoT data with self-adapting, scalable processing. In addition to requiring that all data be streamed into a single central data center, this cloud-centric approach has additional drawbacks that come with centralized data collecting, including a reliance on end-to-end connectivity, higher latency, higher transfer and storage costs, and more.

Combining the Serverless paradigm with data pipelines can be very helpful in avoiding some of the limitations of a cloud-centric approach and in streamlining the design of multi-layer (Cloud, Edge, Fog) IoT applications. Each job in a data pipeline is a process that takes input data, processes it, and then outputs the data. Pipelines are then created by combining these tasks. This makes it easy to integrate basic data processing tasks into more complicated data management services, as well as possibly placing some of the pipeline processes closer to the data sources.

By adopting such strategies, manufacturing systems can handle large-scale, real-time data efficiently while addressing the limitations of centralized data collection. Manufacturing systems' decision-making and quality control procedures can be greatly improved by integrating flexible, multi-layered data pipelines with complex data warehouses like QASDW.

Table 2. 2 Findings of Data Warehouses and Pipeline

No	Tittle	Author	Finding	Gap
1.	AI-Enhanced Data Warehousing: Optimizing ETL Processes for Real-Time Analytics	Hemanth Gadde (2020)	Analysis of traditional ETL limitations and exploration of AI capabilities	Lack of specific real-world examples demonstrating the implementation challenges and long-term sustainability of AI-driven ETL optimizations.
2.	Analyses the Performance of Data Warehouse Architecture Types	Merceedi et al. (2022)	Review and analysis of existing data warehouse architectures; comparison of various research works and practical implementations.	Limited exploration of modern techniques like cloud-based or hybrid data warehouse architectures.
3.	Serverless Data Pipeline Approaches for IoT Data in Fog and Cloud Computing	Poojara et al. (2022)	Simplifies real-time IoT data processing and reduces dependency on centralized cloud solutions.	Reliance on event-driven architecture can be challenging.
4.	ETL Maturity Model for Data Warehouse Systems: A CMMI Compliant Framework	Khan et al., n.d. (2022)	The ETL Maturity Model improves the maturity of ETL processes, enhances data quality, and increases trust in Business Intelligence (BI) systems for better decision-making.	Limited research addressing ETL process maturity in data warehouse systems.
5.	A Comparative Analysis of Data Warehouse Design Methodologies for Enterprise Big Data and Analytics	Asma Qaiser et al. (2023)	Evaluated Inmon, Kimball, Data Vault, and Lambda Architecture for enterprise big data scenarios.	Lacks integration depth across disparate data sources and struggles with high complexity in large datasets.

6.	An Overview of Data Vault Methodology and Its Benefits	Vines & Samoila (2023)	The strengths and weaknesses of Kimball, Inmon, and Data Vault methodologies, emphasizing Data Vault's scalability and agility.	To demonstrate implementation in varying industry contexts, particularly in real-time analytics use cases.
7.	Comparative Analysis of ETL Tools in Big Data Analytics	Qaiser et al., n.d. (2023)	Different ETL tools cater to specific business needs, offering features like real-time processing and scalability.	Lack of a universal tool suitable for all scenarios; complexity in selecting the right tool for specific use cases.
8.	Stream ETL Framework for Twitter-based Sentiment Analysis: Leveraging Big Data Technologies	Ismail et al. (2025)	Proposed a scalable framework that integrates ETL and stream processing for system supports sentiment classification, bias detection, and visualization.	Limited focus on integrating additional advanced mechanisms for stream processing and real-time analytics.

2.3.1 Inmon and Kimball methodologies

William Inmon established an architectural framework for enterprise data warehouses that prioritizes consistent, centralized, and integrated data delivery (Asma Qaiser et al., 2023). His approach involves standardizing, restructuring, and loading data from several operating systems into a reconciled enterprise data warehouse before making it accessible for advanced analytics, dashboards, and management reporting. Among the key ideas and suggestions in the integrated, authoritative data is kept in this single repository, which is arranged for long-term stability and user-friendliness. It makes advantage of enterprise-wide conformance, consistency, and auditable data usage regulations across source systems. Data is ingested, cleaned, conformed, de-duplicated, optimized, and integrated into the reconciled EDW architecture through intricate extract, transform, and load (ETL) operations. This covers data quality assurance, handling errors, and exceptions.

Kimball Method or dimensional modelling approach, Ralph Kimball was a pioneer in the development of dimensional data modelling techniques tailored for business intelligence. It focuses on making integrated enterprise data understandable and accessible to business users through the use of BI tools for dashboards, reporting, and data discovery. Dimensional schemas employ descriptive qualities, hierarchies, and relationships to explain business processes and events in terms that users are familiar with. This is a key idea in the Kimball technique. Quantitative measures are present in facts for analysis. Data profiling, error management, anomaly detection, and metadata tagging from source to target are ways to keep clean and correct data. Document the whole data lineage.

Kimball's architecture that was proposed in 1996 consist in a two-layer model (Vines & Samoila, 2023). Before the raw data is fed into the second tier, the data warehouse duplicated into a staging area where all transformations take place. This strategy will allow end users to connect to a common data warehouse model with all of its dimensions (Vines & Samoila, 2023). Its main benefit is that it can be simple to implement. However, because everything has to be loaded a new, it can be difficult if a second model needs to be built using the same landing data.

Data marts, which are mini-data warehouses from which users can extract the necessary information, are part of the new layer of Inmon's architecture, which has three layers to address the issues previously mentioned (Vines & Samoila, 2023). Additional data processing is also required to generate all of the data marts. The data warehouse and data mart should be kept apart when working with models that have multiple use cases. It also offers scalability for adding new requirements or creating new data marts.

2.3.2 Extract, Transform, and Load (ETL) and Extract, Load, transform (ELT)

The Extract, Transform, and Load (ETL) method is the most common method for collecting data from many sources and integrating it a single information source in a data warehouse (Ismail et al., 2025). Data is extracted from resources such relational databases, spreadsheets, XML files, and flat files, converted to data warehouse standards, and then fed into the warehouse as part of the ETL process. Even though analytics and data warehousing require ETL, not all ETL software programs are made equal. The optimal ETL tool may change based on the use cases and conditions.

In the realm of data warehousing architecture, ETL processes play a crucial role behind the scenes. First, the extraction task is carried out, which includes determining which subset of the source data should be processed further. The targeted sources are searched, read, and the data is extracted. Processing and cleaning the data at the record or schema level requires the transformation task. Normalization, denormalization, reformatting, recalculating, summarizing, combining data from many sources, altering important structures, adding temporal components, managing default values, and more are some of the possible functions that may be included in the transformation (Ismail et al., 2025). The loading task in between tasks is writing the processed data to the specified database or storage.

The extraction phase data is retrieved from a variety of source systems, such as spreadsheets, relational databases, XML files, NoSQL databases, and flat files. This step focuses on identifying and importing the relevant subset of data into a staging area, a temporary storage environment for preprocessing. This stage establishes the basis for efficient transformation and loading by guaranteeing that the extracted data is accurate and complete.

The transformation stage is where raw data is cleaned, processed, and standardized to meet the requirements of the target data warehouse. This involves various tasks such as data cleaning which is correcting errors and removing inconsistencies to improve data quality. Normalization and denormalization for structuring data efficiently for storage and analysis. Data merging for combining data from multiple sources to create a unified view.

Advanced technologies like Machine Learning (ML) and Artificial Intelligence (AI) enhance this phase by automating repetitive tasks and improving the accuracy of transformations. For instance, AI-driven tools can detect anomalies and inconsistencies, reducing error rates by up to 30% while increasing the speed and efficiency of the process (Hemanth Gadde, n.d.).

The loading phase is transferring the converted data into a suitable storage system, like a database or data warehouse. Depending on the use case, the loading process may involve bulk loading for historical data or incremental loading for real-time updates. This phase ensures that data is readily accessible for advanced analytics, such as Online Analytical Processing (OLAP), data mining, or reporting applications (Khan et al., 2022). The design of the loading process is often influenced by the specific analytical needs and frequency of updates required.

Extract, Load, transform (ELT) is a more recent approach that is often used in cloud-based systems and big data environments. ELT excels in more dynamic environments, where data can come in various forms. The data is transformed within the target system, giving analysts the freedom to experiment with different transformations as needed. ELT is often used in real-time analytics environments where it's essential to have fresh data available immediately (Kishore Reddy Gade, 2017). Data can be extracted and loaded into the system in near real-time, and transformations can be applied as required.

For this study ETL which is traditional ETL tools can struggle to scale, particularly when handling big data until be used. The limitations are often due to the computing resources needed to perform the transformations before loading. ETL processes are well-suited for structured data environments where the data schema is consistent. ETL is generally used in batch processing scenarios, which means that it works well with periodic data updates and there is no need for real-time processing. Data warehousing systems that store structured data from various operational systems often rely on ETL processes to ensure that the data is clean, accurate, and properly formatted before it enters the warehouse.

2.4 Data Preprocessing

Data pre-processing is first stage of pre-processing is the process of transforming raw data into a format that will be more effective and usable for processing at the next stage. The minmax technique is used in the initial phases to normalize the data. It is because all training data have the same scale, such as being between 0 and 1, normalization helps shorten training times. This is accomplished by applying, where X is the initial value before to normalization and Xnorm is the result of normalization. In this case, Xmax and Xmin stand for each feature's maximum and minimum values, respectively (Ahmad & Aziz, 2019).

To maximize the information capture and process, data preparation entails changing the data values of a particular dataset. Normalizing the data make the technique used for it related processing less complex because dataset's maximum and minimum usually differ greatly. The normalization of the data provides a sufficient advantage for the categorization of neural network-related algorithms, according to Larriva-Novo et al. (2021). In this instance, normalizing the input values will expedite the training phase and make the neural network more effective if the back-propagation approach is applied.

Table 2. 3 Findings of Data Preprocessing

No	Tittle	Author	Finding	Gap
1.	Data Preprocessing and Feature Selection for Machine Learning Intrusion Detection Systems	Ahmad & Aziz (2019)	The study shows that combining feature selection methods (CFS + PSO) and data preprocessing (normalization and discretization) improves accuracy. Specifically, SVM	The gap in previous research was the lack of optimization for feature selection in intrusion detection systems (IDS), leading to the use of irrelevant features in models.
2.	A Proposed Technique for Conversion of Unstructured Agro-data to Semi-structured or Structured Data	Jui Joshi (2019)	The Couchbase tool provides scalability, performance, and availability for managing agricultural big data, enabling meaningful insights from previously unstructured data.	Existing techniques often work for single text or cluster setups but lack generalizability and efficiency.
3.	Sequence Modeling Approach for Structured Data Extraction from Unstructured Text	Deshmukh & Sengupta, n.d (2019)	Found sequence tagging models to be more effective than machine translation models for extracting structured data from unstructured text. Proposed variants improved performance	Machine translation models were found unsuitable for this task due to limitations in capturing word-level information.
4.	An IoT-Focused Intrusion Detection System Approach Based on Preprocessing Characterization for Cybersecurity Datasets	Larriva-Novo et al. (2021)	Evaluation of preprocessing techniques for machine learning algorithms on cybersecurity datasets in IoT systems.	Lack of standardized, up to date IoT-specific datasets for NIDS (e.g., limited data for emerging IoT attack patterns).

2.4.1 Feature Detection

Finding the best feature in the data set is the goal of feature selection. The data can be categorized using machine learning techniques into a collection of class features and targets. The number of variables or features utilized in machine learning or pattern recognition applications has increased from tens to hundreds. The challenge of eliminating meaningless and irrelevant variables is addressed by a few strategies. Variable elimination, or feature selection, enhances efficiency, lowers processing requirements, lessens dimensional curse effects, and aids in data comprehension (Ahmad & Aziz, 2019).

Feature selection offers several advantages, including lower training and data collection costs, smaller models, better classification performance, and maybe simpler classification models. For feature selection, numerous algorithms have been proposed. Feature selection strategies are frequently divided into three categories in numerous works: filter-based, wrapper, and embed techniques (Lyu et al., 2023). Hybrid approaches and a number of other strategies, however, do not fit into these three groups. An overview of them is given in this section. The filter-based approach is the main topic of this paper due to its many benefits and popularity among developers and researchers. Two crucial technical components of filter-based techniques, search algorithms and relevance measures, are presented in detail in this section after the introduction of filter-based approaches.

2.4.2 Unstructured data to Structured data

Structured data is the data that matches to a specific hierarchical format is referred One type of structured data is the information found in a traditional RDBMS. The search engine can display information more easily when it is stored in a specific format. This covers information found in relational databases and spreadsheets. The kinds of structured data are reports, logs, tags, objects.

The opposite of structured data is unstructured data. Analysing this type of data takes a lot of effort because it lacks organization. The data is divided into two categories and came from either human or machine-generated sources, non-textual unstructured data is multimedia data like MP3 audio files, videos and images. Textual unstructured data examples are email messages, documents, and metadata.

There are many different types of unstructured data, including documents, surveys, reports, logs, and more. Restricting the direct capture of this data in structured form prevents the data from naturally occurring and leaves out important components (Deshmukh & Sengupta, n.d.). However, structured data is easier to understand than a corresponding text since it delivers the data in a clear, straightforward manner. Tables may be created from structured data and saved in databases with efficiency. To get relevant search outcomes, it can be indexed, inquired for, and searched. As a result, structured representation is crucial for enabling machine data consumption. Furthermore, such structured representation is necessary for human consumption in the age of abundant data.

Extraction of structured information from unstructured text has always been a problem of interest for NLP community. Structured data makes it easier for humans and machines to consume and is concise to store, search, and retrieve. Historically, many processing approaches have been used to extract structured data from text by using heuristics and rules relevant to a given topic. In this study, we use the advancements in sequence modelling to the structured data extraction problem (Deshmukh & Sengupta, n.d.).

The implementation of NLP technologies is to work in creating a mechanism to transfer semantics unstructured to SQL data. The author use graph visualization techniques called spring growth to visualize mined structured data with an approach presenting such visualization. The application of the semantics tools will help analyze unstructured data. The information is stored in Resource Description Framework (RDF) knowledge base and queried using SPARQL query language.

The transformation from unstructured to structured data is pivotal for enabling efficient querying and analytics, especially in environments dealing with massive data volumes such as IoT systems or manufacturing pipelines. SQL-based systems like HiveQL offer scalability and flexibility for semi-structured data, while NoSQL systems like MongoDB are more optimized for high-insertion workloads. The study goal of creating an effective data pipeline for automatic quality detection can be directly supported by modern systems that use these technologies to bridge the gap between unstructured raw data and relevant structured data.

2.5 Automated and Real-Time Reporting in IQC

Automation plays a transformative role in Incoming Quality Control (IQC), significantly reducing manual work and minimizing human error. By integrating AI and machine learning, companies can enhance precision and efficiency in detecting defects and anomalies. Automated systems ensure consistency in quality checks and enable faster decision-making. Real-time reporting capabilities complement this automation, allowing immediate feedback and corrective actions. Such systems improve communication across departments and foster a proactive approach to quality management. With the combination of automation and real-time data, IQC becomes an agile and reliable process that aligns seamlessly with industry 4.0 standards.

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Table 2. 4 Findings of Automated and Real-Time Reporting in IQC

No	Tittle	Author	Finding	Gap
1.	Improving the efficiency of small management information system by using VBA.	Kalwar et al. (2021)	Automation of Repetitive Tasks: The study successfully shows that VBA can effectively automate routine, manual tasks within Microsoft Excel, which are common in SMEs for data	lack of efficient, error-free, and cost-effective information management and reporting solutions for Small and Medium Enterprises (SMEs)

			management and reporting.	
2.	Automation of order costing analysis by using Visual Basic for applications in Microsoft Excel.	Muhammad Kalwar (2022)	Manual data entry and calculations are inherently prone to mistakes. A crucial finding was that the VBA automation effectively eliminated calculation errors	When complex calculations done manually in Excel, it becomes extremely time-consuming. Each calculation, data entry, and update takes considerable effort.
3.	Targeting predictors in random forest regression	Daniel Borup et al. n.d. (2023)	using LASSO regularization, improves the model's ability to identify and utilize strong predictors	Performance in high-dimensional settings without an initial weeding out of irrelevant predictors.
4.	Improving the defect detection performance of Incoming Quality Control	de Vent et al., n.d.(2024)	Improving defect detection at the Incoming Quality Control stage by using automated systems and statistical tools. It emphasizes the need for early-stage detection to prevent defects from reaching production.	While the paper discusses automated detection, it lacks integration with real-time reporting tools like Power BI and does not focus on dashboard-driven decision-making.
5.	Power BI in Manufacturing: Strategies for improved production efficiency and Quality Management	Nirmal Pant, n.d. (2024).	How Power BI can be used to streamline reporting and monitoring in manufacturing processes, enabling better visibility of quality metrics and operational performance.	

6.	Power BI for Manufacturing: Real-Time Analytics to boost Efficiency & quality	Technosolutions, (2024).	The benefits of real-time analytics using Power BI, enabling data-driven insights to improve quality and reduce downtime in manufacturing operations.	The application is broad and generalized for manufacturing; it does not specifically address the integration of ETL pipelines or data cleansing processes from IQC inspection records.

The critical issue of inefficient and error-prone manual order costing analysis in small and medium-sized enterprises (SMEs), particularly when conducted using Microsoft Excel is addressed by Muhammad Kalwar (2022). He states that the traditional manual procedure is not only time-consuming, averaging around 22.80 minutes per order for a single article and color combination, but also very vulnerable to human errors due to repetitive calculations and inputting of data. To address these problems, the study recommends and uses a good solution: automating the order costing report through the assistance of Visual Basic for Applications (VBA) in Microsoft Excel. This automation entails the implementation of sophisticated VBA macros such as conditional statements (if/else) and various looping structures (Do while, for loops) to automate completely all the operations previously manually performed by analysts.

Muhammad Kalwar (2022) also created two user forms for macro execution, and they added eight expert worksheets into the automated Excel template. The automated system was tested with an empirical validation of the time study, and it demonstrated a remarkable 85.92% reduction in report preparation time from the manual process. This significant development demonstrates that the VBA-based automation not only decreases the time required for costing analysis by a large margin but also renders the level of errors much lower, thereby enabling even less qualified personnel to generate precise and credible costing reports with improved efficiency.

The system being developed involved the development of sophisticated VBA macros, utilizing conditional statements and various looping constructs, to programmatically automate processes which were manually being done. Easy-to-use interfaces, such as command buttons on the worksheets, were also built to simplify usage. The effectiveness of this automation using

VBA was evaluated using time studies, which consistently demonstrated enormous reductions in the time required to accomplish the tasks. For instance, in a case study, a newly implemented Demand Management System (DMS) that was built using VBA saved 50% of workers' time for data arrangement, management, and storage. The research points out that VBA is an affordable and practical solution for SMEs, whose resources may not be adequate to buy end-to-end Enterprise Resource Planning (ERP) software. With these critical processes automated, SMEs can significantly improve their reporting effectiveness, reduce errors, and eventually increase office-wide productivity overall (Kalwar et al., 2021).

2.5.1 Predictive Modelling for Random Forest Regression

Daniel Borup et al. (n.d) looks into the problem of incorporating a predictor-targeting step for Random Forest (RF) regression. They are concerned about the fact that in high-dimensional and sparse data environments, this standard use of RF could basically lower the predictive power of the model given that many weak or irrelevant predictors lurk at large.

The highlight of the work is, therefore, the proposition and exhaustive study of the Targeted Random Forest or TRF approach, which ensures an initial dimension reduction step via LASSO regularization, aimed at weeding out less important predictors before feeding the data into the RF algorithm. The selection of the number of predictors to target is then regulated by the Bayesian Information Criterion (BIC), thus instilling in the procedure a data-driven nature.

The study brings to light several important results. First, from a theoretical perspective, they show how targeting of predictors directly affects the probability of an RF tree splitting along strong (informative) predictors. Then the simulations show that this pre-screening step pushes upwards the lower bound on this probability by reducing largely the estimation error on the CART impurity decrease.

In summary, the Random Forest regression model's application for predicting the PPM defect rate from past IQC data was the exclusive focus of this investigation. Because of its resilience, capacity to manage nonlinear interactions, and efficiency in handling datasets with numerous variables without requiring a great deal of preprocessing, Random Forest was selected. This study sought to assess the regular Random Forest model as a baseline for predictive modeling in a manufacturing quality control environment, despite the existence of advanced approaches such as Targeted Random Forest (TRF) to enhance performance in high-dimensional settings. The findings show that Random Forest produces accurate forecasts, which makes it a good option for IQC's initial data-driven quality forecasting implementation.

2.5.2 Real-Time Visualization with Power BI

Power BI has emerged as a leading tool for real-time data visualization in IQC processes. Its ability to consolidate complex datasets into interactive dashboards allows stakeholders to track performance metrics efficiently. Create interactive dashboards that display live data on production rates, machine performance and defect rates. The benefits is timely intervention can prevent bottlenecks and reduce downtime (Nirmal Pant, n.d.). Literature supports the use of Power BI for daily reports, weekly trends, and monthly summaries, emphasizing its versatility in meeting diverse reporting needs. Real-time data production lines allow manufacturers to refine their processes on the fly. With Power BI in manufacturing, companies can monitor key performance indicators (KPIs), track equipment status and swiftly identify bottlenecks (Technosolutions, 2024).

The integration of visual tools such as charts and graphs in quality monitoring fosters stakeholder engagement, enabling informed decision-making. The objective is aggregate data from various sources into a unified platform. The data sources is integrate data from ERP system, production machines, quality control databases and supply chain management tools (Nirmal Pant, n.d.). Several case studies illustrate the effective use of Power BI in monitoring defect rates and identifying patterns, underscoring its value in promoting proactive quality control. Power Bi is at the center of the manufacturing sector's quick adoption of digital transformation. Manufacturers can improve their decision-making, increase quality control, and streamline operations by utilizing real-time analytics (Technosolutions, 2024). Manufacturing will be intelligent, efficient, and flexible in the future thanks to power BI, not just data driven.

2.6 Summary

In conclusion, Chapter 2 provides an extensive review of the literature relevant to the development of an automated quality system for manufacturing operations. This chapter establishes a theoretical framework and identifies key methodologies and tools that will inform the design and implementation of the proposed quality detection system, setting the stage for the subsequent methodological development in Chapter 3. The literature review also explores from comprehensive review of related to develop of an automated quality database he reviews covers essential concepts such as quality control frameworks, data integration techniques, and the application of ETL processes to automate the consolidation of inspection data into a centralized database. Additionally, the literature explores the use of predictive modelling in quality assurance, with a focus on classification techniques that can forecast the likelihood of material lot based on historical inspection results. To summarize, this chapter highlights the importance of combining ETL automation, predictive modelling, and interactive data visualization to improve IQC processes.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter explains the research methods used to develop the automated quality monitoring system for TT Electronics. The focus is on improving data processing, building a reliable data pipeline, and setting up a system to monitor quality more effectively. The approach follows Industry 4.0 principles, which promote automation, data-based decisions, and smart analytics. The methodology is also guided by the DMAIC framework (Define, Measure, Analyze, Improve, Control), which helps to structure the problem-solving process. In addition, the data science life cycle is used to support each phase, covering planning, understanding business needs, selecting analysis methods, and collecting and preparing data for accurate insights.

3.2 Research Design

Research framework in figure 3.1 is the process begins with identification and collection of data, which is structured dataset. This to ensure that all the data is collected for analysis and processing. The next step is data extraction that involves getting raw data and converting it into usable data format. The extracted data is then moved to the data preprocessing stages, where cleaning, transforming, and exploratory data analysis (EDA) is done. This step is crucial for addressing issues like missing values, duplicate records, and inconsistent formats, while also uncovering patterns and insights within the data.

After the preprocessing, the data integrated into a data warehouse, which serves as a central respository for storage, querying and quality monitoring. This intergration ensures that the data is well structured and ready for addtional analysis. A data pipeline is the essential to automate the flow from source to the warehouse and guarantee effiectiveness updates as new data became available. This complete the secound objective of building ETL pipeline.

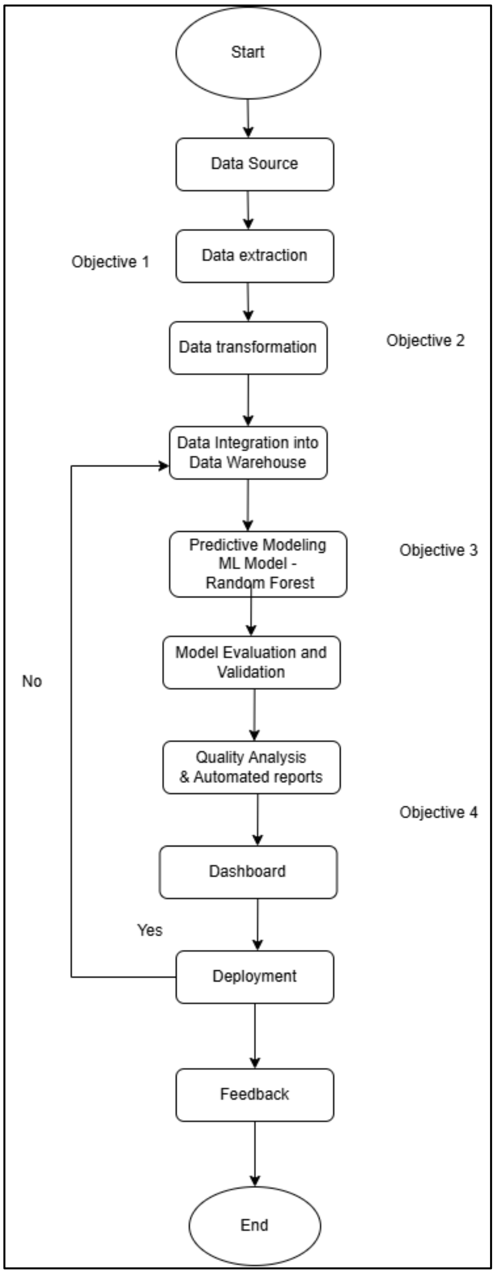


Figure 3. 1 Research Framework

In the data storage phase, the clean and structured data is loaded into Excel based centralized database. This database act as core respository for all inspections data, enabling efficient query processing and report generation. Then once the data is store, quality analysis and reporting are conducted. The automated report generation are implemented using VBA to achived objective 1.

The system incorporates the creation of a predictive model in addition to reporting. In order to forecast the PPM (Parts Per Million) defect rate of incoming material lots, this model which was constructed using the Random Forest algorithm is trained on past inspection data. Following data storage, the system moves on to the automated report production and quality analysis stage, where inspection findings, trends, and performance metrics are displayed in real time via Power BI dashboards.

If any part of the process, such as the pipeline, does not meet the desired standards, the framework includes iterative feedback loops that return to earlier stages like data preprocessing or system development for refinement. Following a successful implementation, the system enters the deployment phase, when end users can access and use it. At this stage, continuous monitoring and feedback collection are critical for evaluating system performance.

The Turtle Diagram shown in Figure 3.2 is a straightforward drawing that helps to understand any work process, big or small. Its major function being displaying all crucial aspects of a task on a single sheet for easy understanding as it explains what the main task is, what tools and persons are involved, what information or objects come into it, and what comes out from it. It explains in crucial ways how the job will be done and how we shall measure if the job was done successfully. Just think of it as a clear process road map that lets everyone involved grasp the big picture quickly so that they can effectively plan on how to improve it. This study focuses on creating a Database and Dashboard to automate Incoming Quality Control (IQC). Resources used will include Excel, Python, Power BI, and the standard Laptop/PC. The core team for execution will comprise Maisarah, Pn Zakiyah, Dr Shaimel, and Dr Zanariah, while the procedure will start with Inputs taken from the current IQC Database, MR files, and IQC Checklists. From these Inputs, through automation using VBA Excel and analysis by a Random Forest model, and with dashboard development using Power BI, would generate Outputs: Predicted PPM (Parts Per Million), a fully Automated Database, and Dashboard Visualizations. Subsequently, the Effectiveness of the entire system would be measured by comparing predicted vs. actual productivity profiled over time by cross-checking the new automated database against its original data sources to ensure accuracy and reliability.

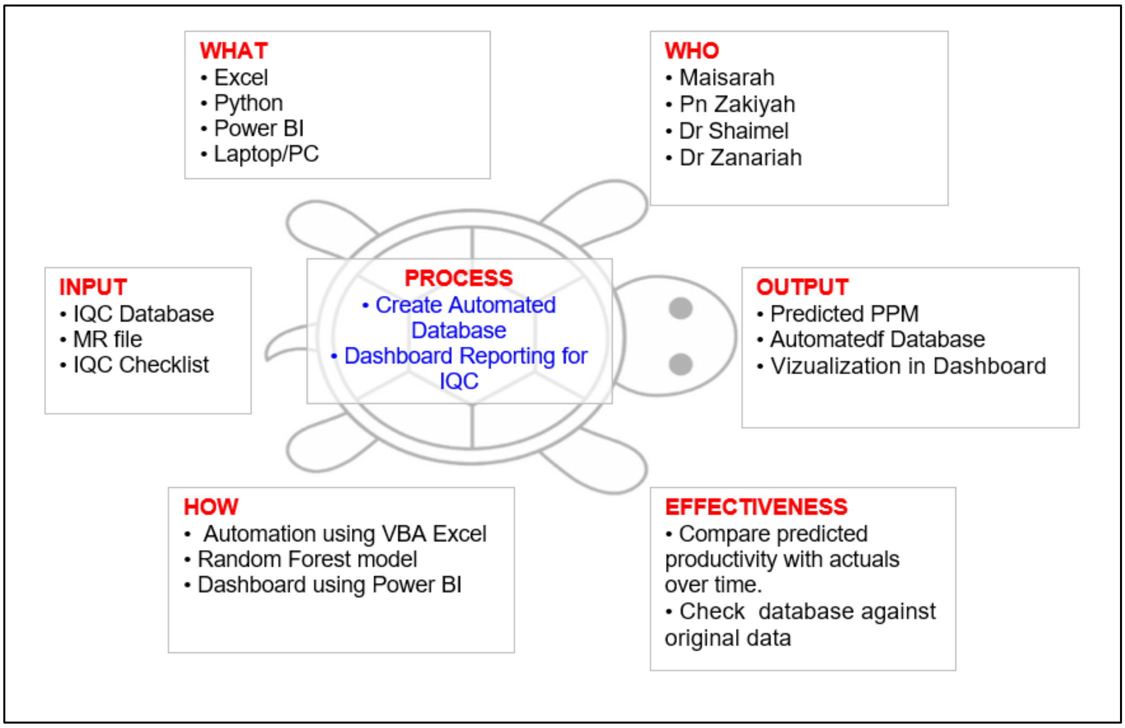


Figure 3. 2 Turtle Diagram

Meanwhile, the DMAIC approach is a structured methodology widely used for quality improvement, especially in Six Sigma projects. DMAIC stands for Define, Measure, Analyse, Improve, and Control. Each phase plays a critical role in identifying problems, analysing root causes, implementing effective solutions, and maintaining long-term improvements.

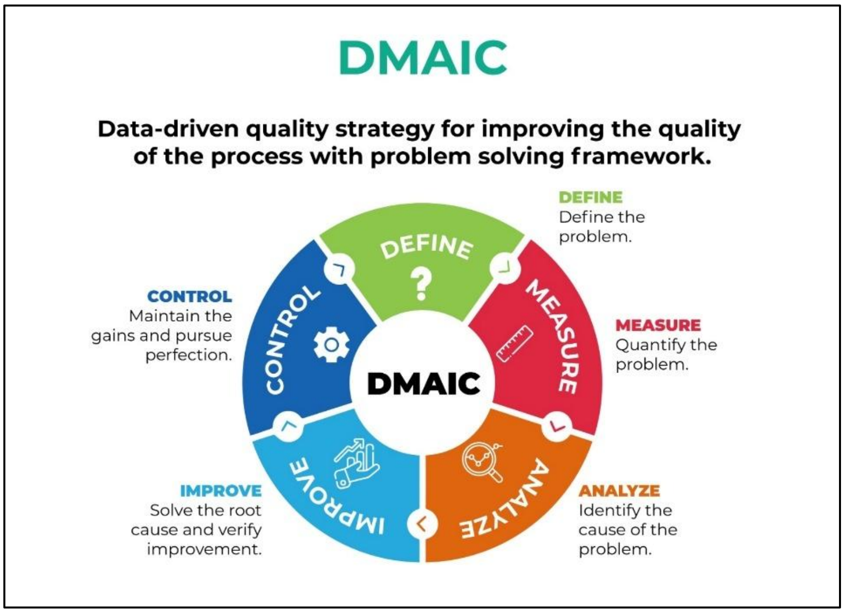


Figure 3. 3 Figure of DMAIC Cycle

In this study, the DMAIC framework serves as a systematic guide to develop an automated quality data monitoring system for Incoming Quality Control (IQC) in the manufacturing environment. By following each phase step-by-step, the study ensures that the solutions are data-driven, efficient, and aligned with organizational goals.

3.3 Define Phase

During the Define phase, the primary issues in the current IQC process are recognized. This encompasses problems like excessive dependence on manual data input, absence of integration among data sources, and lagging reporting. The study aims are explicitly outlined, highlighting the necessity to automate data gathering, standardize inspection documentation, and enhance decision-making via real-time dashboards.

3.3.1 Business Understanding

The manufacturing industry, particularly with regard of TT Electronics is following high-quality standards in production to sustain competitive advantages. Quality assurance (QA) is important in ensure that before any customer delivery product meet the expected quality standards. However, TT Electronics faces difficulties to accomplish this because the lack of an integrated data pipeline for quality control procedures, leading to significant reliance on manual processes for detecting quality issues, which is time-consuming and prone to human error. In the quality control process Machine Learning (ML) using Random Forest Regressor to improve accuracy of defect detection, to improve efficiency in handling complex data patterns, and reduce human intervention. To support TT Electronics' transition towards Industry 4.0 by incorporating modern technologies such as the Internet of Things (IoT) to streamline manufacturing processes and maintain a competitive edge.

3.3.2 Analytic Approach

This study utilizes a analytics framework descriptive, diagnostic predictive and prescriptive Analytic so assist in creating an automated quality data monitoring system for Incoming Quality Control (IQC) within the manufacturing setting. Descriptive analytics helps comprehend the present quality performance by examining historical IQC inspection data to uncover trends, patterns, and anomalies. This offers a transparent summary of how inspection outcomes have evolved over time. Diagnostic analytics investigates the underlying causes of quality problems by analyzing crucial factors like supplier performance, material type, and categories of defects. By recognizing the connections among these elements, the system can emphasize crucial areas requiring focus. Predictive analytics entails utilizing machine learning models like Random Forest (RF) to anticipate upcoming quality patterns, including defect rates or Parts Per Million (PPM). These models are chosen for their precision, reliability, and capacity to manage intricate inspection data. Ultimately, prescriptive analytics offers practical suggestions to enhance IQC performance, including optimizing inspection strategies, modifying sample sizes, or focusing on suppliers with higher risk. Collectively, these four tiers of analytics facilitate a data-oriented strategy for quality control, fostering ongoing enhancement and synchronizing the IQC process with Industry 4.0 standards.

3.3.3 Data Requirement

The study will require several types of data to guarantee comprehensive analysis and accurate outcomes. The framework is made up of production data, which includes statistics that reveal the effectiveness and dependability of manufacturing processes of Incoming Quality Control (IQC), such defect rates, Parts Per Million (PPM), and Lot Acceptance Rate (LAR). The data will be collected across multiple dimensions, including supplier name, material, inspection date, inspection status and defect category. This multi-dimensional approach make deeper analysis and traceability through the quality control process.

1) Lar is calculated as:

$$LAR = \frac{TOTAL\ LOT\ INSPECT - TOTAL\ LOT\ REJECTED}{TOTAL\ LOT\ INSPECTED} \tag{3.1}$$

LAR tells us how many lots passed inspection compared to the total lots inspected. If more lots are accepted, the LAR is higher, which is good.

2) PPM is defined as:

$$PPM = \frac{TOTAL\ QUANTITY\ REJECT}{TOTAL\ LOT\ SIZE} X 1{,}000{,}000 \qquad (3.2)$$

PPM tells that how many defective parts exist in every 1 million parts produced. Lower PPM is better because it means fewer defects.

It structed inspection data, the system will also utilize semi-structured data, such as manually enter inspection form or upload excel files. This data format requires preprocessing and transformation to be standardized for analysis. To handle this, ETL pipeline will be developed to clean, organize and unify the data from various sources into a centralized database. The data collected will span across multiple dimensions and the inspection result in the Power BI interactive dashboard enabling real time monitoring and supporting data driven decision making. These dashboards will help track quality trend, identify recurring issues an respond quickly to potential quality risk.

3.4 Measure Phase

The Measure phase focuses on collecting relevant data from various sources such as inspection reports, supplier records, and production data. Key performance indicators (KPIs) like defect rate, PPM (Parts Per Million), and LAR (Lot Acceptance Rate) are identified and used to assess the current performance of the IQC process. This phase provides a data baseline to support future analysis.

3.4.1 Data collection

To build a complete and reliable Incoming Quality Control (IQC) report database, collected data from multiple sources to ensure accuracy and comprehensiveness. The source includes the material master list which provides information on raw material, material request (MR) which is help track material flow and usage, IQC checklist and the Notification of Defect

(NOD) report to document identified defect. Additionally, gathered data from the SAP system to align with existing data. Beyond these documents, conducted direct inspection of IQC item. This involved checking item that showed potential quality issues and performing sampling based on the right format to ensure valid results. Defective item found during these inspection wad recorded and analysed. This through data collection process is an important first step before proceeding to data analysis. Completing and verifying the database will enable the creation of an automated, data-driven quality monitoring system that supports accurate reporting and continuous improvement in TT Electronics' manufacturing operations.

3.4.2 Data preparation

Data preparation involve cleaning and transforming the raw data into format suitable for data analysis. The extract phase determines where the data from the various sources, which can include dataset and manual inputs. The retrieve data its original state, whether structured in table such as inspection report and material log, as well as manual input record by inspectors. The data can be scheduled by set up periodically data pulls in real time, hourly and daily to keep the dataset updated. The challenges of data preparation are inconsistent format across sources and handling large volume of data efficiently.

The transformation phase involves reformatting, cleaning, and enhancing the extracted data to make it consistent and usable. This stage may include removing or imputing missing values, correcting inconsistencies, handling outliers, and standardizing or normalizing variables. The loading phase involves moving the transformed data into a target system where it can be analysed. Once cleaned and standardized the data is loaded into a centralized data warehouse for analysis and visualization tools like Powe BI. It enables real time reporting, monitoring of quality KPIs and generate of actionable insights. Data preparation using ETL ensures that the data quality is accurate, complete, and consistent because its scalability to prepares the data for immediate use in visualization and decision making.

3.4.3 Data understanding

Data understanding is the phase where the collected data is explored and examined to gain insight into its structure, quality, and relevance to the problem. The type of data characterized used in the research is both quantitative and qualitative attributes, such as numeric values for sample sizes, rejection quantities, inspection durations, and calculated KPIs like Parts Per Million (PPM) and Lot Acceptance Rate (LAR). The data from IQC report, material and inspection report completed by IQC inspector. The dataset is yearly of historical data and real-time data where the amount of past data available for the analysis.

Table 3. 1 Shown variable of the dataset

Attributes	**Qualitative/ Quantitative**	**Data Type**	**Description**
Supp Name	Qualitative	Text/String	Name of the supplier.
Customer	Qualitative	Text/String	The customer receiving the product.
Lot Size	Quantitative	Integer	Total quantity in the lot received.
Disposition	Qualitative	Text/String	Final decision (e.g., Accept, Reject, Rework).
Tot Qty Reject	Quantitative	Integer	Total number of units rejected.
MR Date	Quantitative	Date	Date the MR was raised.
Date Inspected	Quantitative	Date	Date the inspection was performed.
Completed Insp Date	Quantitative	Date	Date the inspection process was completed.
Total Lot Inspected	Quantitative	Integer	Number of lots inspected in total.
Total Lot Rejected	Quantitative	Integer	Number of lots rejected.
Total Lot Size	Quantitative	Integer	Total size (quantity) across all inspected lots.
PPM	Quantitative	Float	Parts per million defect rates.

LAR	Quantitative	Float	Lot Acceptance Rate, typically shown as a percentage.

3.4.4 ETL Pipeline

The Extract, Transform, Load (ETL) pipeline is a crucial component of data integration that enables businesses to streamline the flow of data various sources into a centralized database, enabling seamless processing and analysis. The study focuses on developing an ETL pipeline to handle production related data by automating the system.

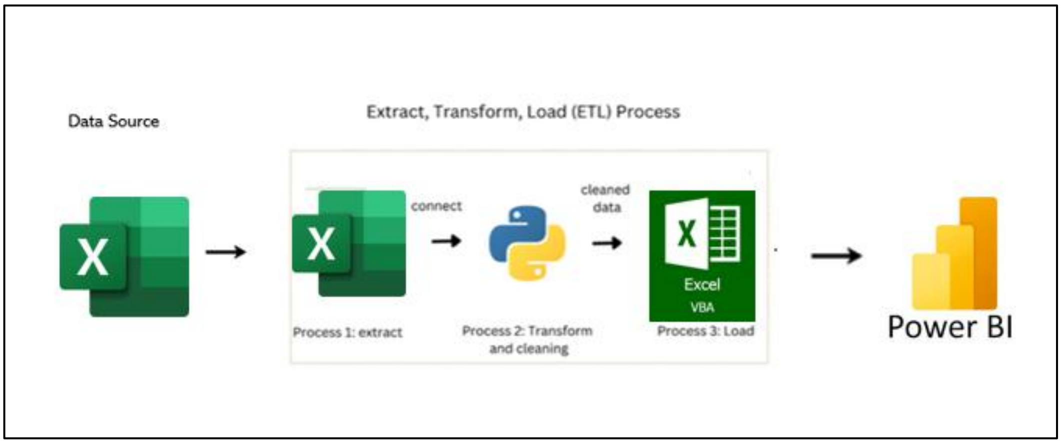


Figure 3. 4 ETL pipeline structure

The data pipeline is the backbone of the data integration process that data flows from different sources to a central database. This project has three stages extraction, transformation, and loading. Extraction phase involves pulling out data from various sources such as Excel files, databases, and API responses. Data is in the form of Pandas Data frames representing structured datasets. Data is then transformed to fit the schema of the destination database as well as improve its quality. The loading stage involves inserting the processed data into the centralized Excel database, facilitated by VBA (Visual Basic for Applications), which automates the structured integration of incoming data. The function takes care of any errors during the loading process. The ETL Pipeline serves as an efficient method for transferring software. It can do extraction and manipulation with Pandas.

In the extract step of the system, it gathers information from different sources, such as Excel spreadsheets submitted by inspectors, and shared quality reports from suppliers. In instances where semi-structured files are used, Python parsing tools like the pandas and openpyxl libraries are used to systematically extract tabular data from Excel files.

Transform data is to intensive pre-processing to ensure accuracy and consistency. Data cleansing is performed, removing data duplicates and fixing missing values through standardization of formats such as defect codes and date formats, all these done by use of version libraries such as pandas and NumPy. Important collected features include product ID, defect type, severity level, and inspection date. Trends of derived metrics like defect rates, reworks frequencies, and production batch trends are formulated to depict areas of concern. Highly elevated detection methods enable pattern detection, allowing data analysis techniques to express abnormal trends, spikes in defect occurrence, or discrepancies in batch performance.

The final stage, data loading, involves systematically inserting the fully processed and validated data into a centralized Excel database. This loading process is automated and managed through VBA, which facilitates efficient and structured integration of cleaned data into the designated Excel repository, thereby maintaining a dynamic and accessible database. Processed data is also fed into visualization tools such as Power BI for creating dynamic dashboards with critical insights like visible defect trends or abnormal patterns and production performance measures. Overall, such visualization permits real-time monitoring of quality issues, allowing stakeholders at TT Electronics to identify problem areas, activate corrective action, and further develop operational efficiency.

This ETL pipeline takes good care of the management, processing, and visual effect of quality assurance data so that the pipeline is accurate in detecting abnormal patterns, thereby enhancing production quality management.

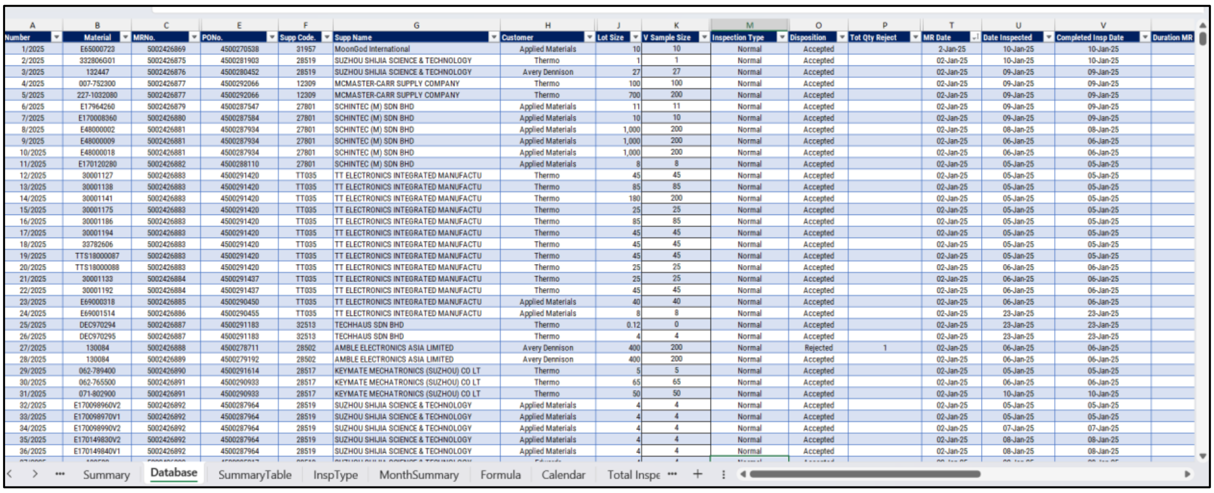
3.5 Analyse Phase

During the Analyse stage, the gathered data is to see uncover trends, patterns, and underlying causes of quality problems. The aim is to identify the reasons behind specific defects, determine which suppliers lead to low quality, and find out which materials or products are most likely to fail. Methods like Pareto analysis, correlation analysis, and visual inspection trends can be employed to back up conclusions.

3.5.1 Data Collection and Understanding

First step involves collecting inspection data from multiple sources, including IQC Checklist, Material Report, and Reject Report. These datasets were originally provided in Microsoft Excel format and covered different product categories such as magnetic parts (MAG), electronic manufacturing services (EMS), plastic parts, and metal components.

The dataset contains important information for the analysis the data collected from IQC checklist, Material report and reject report. The original dataset obtains in Microsoft Excel workbook format. For Incoming Quality Report, there is database sheet in the file, which indicates that its manufacturing process started. The IQC dataset is still ongoing data in this manufacturing process and all the data collected are manual handling. This is the dataset information:



A	B	C	E	F	G	H	J	K	M	O	P	T	U	V	
Number	Material	MRNo.	PONo.	Supp Code.	Supp Name	Customer	Lot Size	V Sample Size	Inspection Type	Disposition	Tot Qty Reject	MR Date	Date Inspected	Completed Insp Date	Duration MR
1/2025	E65000723	5002426869	4500270538	31957	MoonGod International	Applied Materials	10	10	Normal	Accepted		2-Jan-25	10-Jan-25	10-Jan-25	
2/2025	332806G01	5002426875	4500281903	28519	SUZHOU SHIJIA SCIENCE & TECHNOLOGY	Thermo	1	1	Normal	Accepted		02-Jan-25	10-Jan-25	10-Jan-25	
3/2025	132447	5002426876	4500280452	28519	SUZHOU SHIJIA SCIENCE & TECHNOLOGY	Avery Dennison	27	27	Normal	Accepted		02-Jan-25	09-Jan-25	09-Jan-25	
4/2025	007-752300	5002426877	4500292066	12309	MCMASTER-CARR SUPPLY COMPANY	Thermo	100	100	Normal	Accepted		02-Jan-25	09-Jan-25	09-Jan-25	
5/2025	227-1032080	5002426877	4500292066	12309	MCMASTER-CARR SUPPLY COMPANY	Thermo	700	200	Normal	Accepted		02-Jan-25	09-Jan-25	09-Jan-25	
6/2025	E17964260	5002426879	4500287547	27801	SCHINTEC (M) SDN BHD	Applied Materials	11	11	Normal	Accepted		02-Jan-25	09-Jan-25	09-Jan-25	
7/2025	E170008360	5002426880	4500287584	27801	SCHINTEC (M) SDN BHD	Applied Materials	10	10	Normal	Accepted		02-Jan-25	09-Jan-25	09-Jan-25	
8/2025	E48000002	5002426881	4500287934	27801	SCHINTEC (M) SDN BHD	Applied Materials	1,000	200	Normal	Accepted		02-Jan-25	08-Jan-25	08-Jan-25	
9/2025	E48000009	5002426881	4500287934	27801	SCHINTEC (M) SDN BHD	Applied Materials	1,000	200	Normal	Accepted		02-Jan-25	06-Jan-25	06-Jan-25	
10/2025	E48000018	5002426881	4500287934	27801	SCHINTEC (M) SDN BHD	Applied Materials	1,000	200	Normal	Accepted		02-Jan-25	06-Jan-25	06-Jan-25	
11/2025	E170120280	5002426882	4500288110	27801	SCHINTEC (M) SDN BHD	Applied Materials	8	8	Normal	Accepted		02-Jan-25	05-Jan-25	05-Jan-25	
12/2025	30001127	5002426883	4500291420	TT035	TT ELECTRONICS INTEGRATED MANUFACTU	Thermo	45	45	Normal	Accepted		02-Jan-25	05-Jan-25	05-Jan-25	
13/2025	30001138	5002426883	4500291420	TT035	TT ELECTRONICS INTEGRATED MANUFACTU	Thermo	85	85	Normal	Accepted		02-Jan-25	05-Jan-25	05-Jan-25	
14/2025	30001141	5002426883	4500291420	TT035	TT ELECTRONICS INTEGRATED MANUFACTU	Thermo	180	200	Normal	Accepted		02-Jan-25	05-Jan-25	05-Jan-25	
15/2025	30001175	5002426883	4500291420	TT035	TT ELECTRONICS INTEGRATED MANUFACTU	Thermo	25	25	Normal	Accepted		02-Jan-25	05-Jan-25	05-Jan-25	
16/2025	30001186	5002426883	4500291420	TT035	TT ELECTRONICS INTEGRATED MANUFACTU	Thermo	85	85	Normal	Accepted		02-Jan-25	05-Jan-25	05-Jan-25	
17/2025	30001194	5002426883	4500291420	TT035	TT ELECTRONICS INTEGRATED MANUFACTU	Thermo	45	45	Normal	Accepted		02-Jan-25	05-Jan-25	05-Jan-25	
18/2025	33782606	5002426883	4500291420	TT035	TT ELECTRONICS INTEGRATED MANUFACTU	Thermo	45	45	Normal	Accepted		02-Jan-25	05-Jan-25	05-Jan-25	
19/2025	TTS18000087	5002426883	4500291420	TT035	TT ELECTRONICS INTEGRATED MANUFACTU	Thermo	45	45	Normal	Accepted		02-Jan-25	05-Jan-25	05-Jan-25	
20/2025	TTS18000088	5002426883	4500291420	TT035	TT ELECTRONICS INTEGRATED MANUFACTU	Thermo	25	25	Normal	Accepted		02-Jan-25	06-Jan-25	06-Jan-25	
21/2025	30001133	5002426884	4500291437	TT035	TT ELECTRONICS INTEGRATED MANUFACTU	Thermo	25	25	Normal	Accepted		02-Jan-25	06-Jan-25	06-Jan-25	
22/2025	30001192	5002426884	4500291437	TT035	TT ELECTRONICS INTEGRATED MANUFACTU	Thermo	45	45	Normal	Accepted		02-Jan-25	06-Jan-25	06-Jan-25	
23/2025	E69000318	5002426885	4500290450	TT035	TT ELECTRONICS INTEGRATED MANUFACTU	Applied Materials	40	40	Normal	Accepted		02-Jan-25	06-Jan-25	06-Jan-25	
24/2025	E69001514	5002426886	4500290455	TT035	TT ELECTRONICS INTEGRATED MANUFACTU	Applied Materials	8	8	Normal	Accepted		02-Jan-25	23-Jan-25	23-Jan-25	
25/2025	DEC970294	5002426887	4500291183	32513	TECHHAUS SDN BHD	Thermo	0.12	0	Normal	Accepted		02-Jan-25	23-Jan-25	23-Jan-25	
26/2025	DEC970295	5002426887	4500291183	32513	TECHHAUS SDN BHD	Thermo	4	4	Normal	Accepted		02-Jan-25	23-Jan-25	23-Jan-25	
27/2025	130084	5002426888	4500278711	28502	AMBLE ELECTRONICS ASIA LIMITED	Avery Dennison	400	200	Normal	Rejected	1	02-Jan-25	06-Jan-25	06-Jan-25	
28/2025	130084	5002426889	4500279192	28502	AMBLE ELECTRONICS ASIA LIMITED	Avery Dennison	400	200	Normal	Accepted		02-Jan-25	06-Jan-25	06-Jan-25	
29/2025	062-789400	5002426890	4500291614	28517	KEYMATE MECHATRONICS (SUZHOU) CO LT	Thermo	5	5	Normal	Accepted		02-Jan-25	05-Jan-25	05-Jan-25	
30/2025	062-765500	5002426891	4500290933	28517	KEYMATE MECHATRONICS (SUZHOU) CO LT	Thermo	65	65	Normal	Accepted		02-Jan-25	06-Jan-25	06-Jan-25	
31/2025	071-802900	5002426891	4500290933	28517	KEYMATE MECHATRONICS (SUZHOU) CO LT	Thermo	50	50	Normal	Accepted		02-Jan-25	10-Jan-25	10-Jan-25	
32/2025	E170098960V2	5002426892	4500287964	28519	SUZHOU SHIJIA SCIENCE & TECHNOLOGY	Applied Materials	4	4	Normal	Accepted		02-Jan-25	05-Jan-25	05-Jan-25	
33/2025	E170098970V1	5002426892	4500287964	28519	SUZHOU SHIJIA SCIENCE & TECHNOLOGY	Applied Materials	4	4	Normal	Accepted		02-Jan-25	05-Jan-25	05-Jan-25	
34/2025	E170098990V2	5002426892	4500287964	28519	SUZHOU SHIJIA SCIENCE & TECHNOLOGY	Applied Materials	4	4	Normal	Accepted		02-Jan-25	07-Jan-25	07-Jan-25	
35/2025	E170149830V2	5002426892	4500287964	28519	SUZHOU SHIJIA SCIENCE & TECHNOLOGY	Applied Materials	4	4	Normal	Accepted		02-Jan-25	08-Jan-25	08-Jan-25	
36/2025	E170149840V1	5002426892	4500287964	28519	SUZHOU SHIJIA SCIENCE & TECHNOLOGY	Applied Materials	4	4	Normal	Accepted		02-Jan-25	08-Jan-25	08-Jan-25	

Summary | Database | SummaryTable | InspType | MonthSummary | Formula | Calendar | Total Inspe

Figure 4. 1 IQC Database Table

In the dataset the sheet like in figure 4.1 contains all the group of data entries with all the data from IQC like material Name, MR Number, Date of Inspection, Reject Quantity, LAR (Lot Acceptance Rate), PPM (Parts Per Million), and their respective target values. As the data was collected manually by IQC personnel, human error and variability in data entry was observed, necessitating a structured preprocessing step.

To build the database, data is extracted from two different daily files such as the MR file and the IQC Checklist. Examples of these daily files are shown in Figure 4.2.

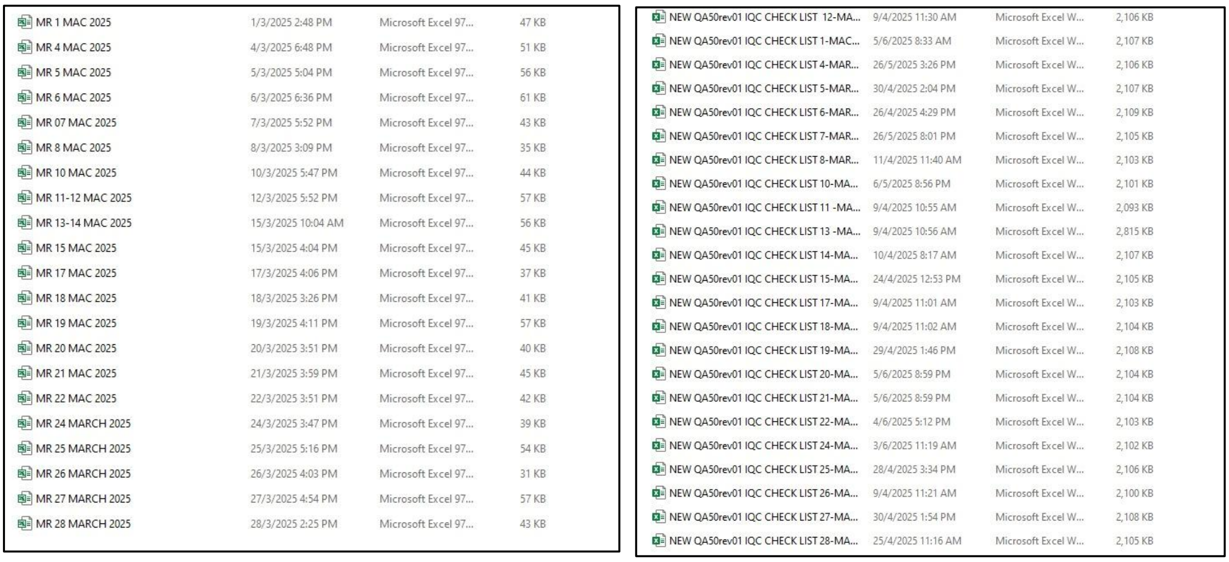


Figure 4. 2 Folder Daily IQC that contain MR file and the IQC Checklist.

3.5.2 Data Preparation

Data preparation is critical phase in ensure raw data inspection data is reliable, consistent and ready for analysis and visualization. The raw data through preparation for this reporting processing and visualization, the raw dataset we cleaned, standardized, and transformed. This phase addressed various data quality issues such as inconsistency, missing value and duplicate records. Column headers standardized across all datasets to ensure uniformity.

The Figure 4.2 shows the output of the .info() method from a pandas DataFrame, which provides a concise summary of the dataset structure. This summary includes the number of entries, the range index, column names, the number of non-null (non-missing) values in each column, the data type for each column, and overall memory usage.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5272 entries, 0 to 5271
Data columns (total 30 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   Material                     5272 non-null   object
 1   MRNo.                        5272 non-null   int64
 2   PONo.                        5272 non-null   int64
 3   Supp Code.                   5272 non-null   object
 4   Supp Name                    5272 non-null   object
 5   Customer                     5272 non-null   object
 6   Description                  5272 non-null   object
 7   Lot Size                     5272 non-null   float64
 8   V Sample Size                5272 non-null   float64
 9   Inspection Type              4976 non-null   object
 10  Disposition                  5268 non-null   object
 11  Tot Qty Reject               29 non-null     float64
 12  MR Date                      5272 non-null   datetime64[ns]
 13  Date Inspected               5211 non-null   datetime64[ns]
 14  Completed Insp Date          5211 non-null   datetime64[ns]
 15  Duration MR to Complete      5270 non-null   object
 16  Duration Inspect to Complete 5270 non-null   object
 17  WW                           5272 non-null   int64
 18  Total Lot Inspected          5272 non-null   int64
 19  Total Lot Rejected           5272 non-null   int64
 20  Total Lot Size               5272 non-null   float64
 21  Total Quantity Rejected      5272 non-null   int64
 22  Total V Sample Size          5272 non-null   float64
 23  Total M Sample Size          5272 non-null   int64
 24  PPM                          5271 non-null   float64
 25  LAR                          5200 non-null   float64
 26  Target PPM                   5272 non-null   int64
 27  Target LAR                   5272 non-null   float64
 28  Month                        5211 non-null   object
 29  Remark                       60 non-null     object
dtypes: datetime64[ns](3), float64(8), int64(8), object(11)
memory usage: 1.2+ MB
```

Figure 4. 3 Data information

```
#check missing value
data.isnull().sum()
```

Material	0
MRNo.	0
PONo.	0
Supp Code.	0
Supp Name	0
Customer	0
Description	0
Lot Size	0
V Sample Size	0
Inspection Type	296
Disposition	4
Tot Qty Reject	5243
MR Date	0
Date Inspected	61
Completed Insp Date	61
Duration MR to Complete	2
Duration Inspect to Complete	2
WW	0
Total Lot Inspected	0
Total Lot Rejected	0
Total Lot Size	0
Total Quantity Rejected	0
Total V Sample Size	0
Total M Sample Size	0
PPM	1
LAR	72
Target PPM	0
Target LAR	0
Month	61
Remark	5212

MRNo.	0
PONo.	0
Supp Code.	0
Supp Name	0
Customer	0
Description	0
Lot Size	0
V Sample Size	0
Inspection Type	0
Disposition	0
Tot Qty Reject	0
MR Date	0
Date Inspected	59
Completed Insp Date	59
Duration MR to Complete	0
Duration Inspect to Complete	0
WW	0
Total Lot Inspected	0
Total Lot Rejected	0
Total Lot Size	0
Total Quantity Rejected	0
Total V Sample Size	0
Total M Sample Size	0
PPM	0
LAR	0
Target PPM	0
Target LAR	0
Month	0
Remark	0

Figure 4. 4 Checking missing values and after imputation

The figure above shows that comparison of missing values in dataset before and after data cleaning. On the column have missing values on lot quantities, reject, date inspection, LAR and remark. This missing value can effect the quality of analysis, so they nedd to handled carefully. The next show dataset afrer imputation techniques which means the missing valyes have been filled in. This may been done by using defaukt values, average or by calculating value from related column. As a result, all columns now show zero missing values except for data inspection and complete inspection date because of the data is inprogress entreis, indicate that the dataset is clean and ready for further analysis. The data going forward an ETL Pipeline using Python.

3.5.3 System Design and Architecture

The system design and architecture are a technical overview of the automated quality data monitoring and reporting system developed for TT Electronics. The system's design integrates a strong data pipeline, centralized storage, and visualization tools to manage the entire data flow process, from collecting data to real-time reporting. This architecture supports automation, scalability, and continuous monitoring in quality control procedures, all of which are in line with Industry 4.0 concepts.

The first component is the data input layer which collects structured data from some sources, such as Excel inspection forms and materials report from Incoming Quality Control (IQC) reports. These Excel-based input forms are automated using VBA, which streamlines data entry and ensures consistency. The data inputs are passed into ETL (Extract, Transform, Load) engine, which is developed using Python libraries like Pandas and NumPy, cleanses and standardizes the data before it is loaded directly into the central Excel database. This step extract of raw data, transformation through data cleaning and standardization and loading into a centralized database.

Once processed, the data is stored in the centralized Excel database, which serves as the main repository for all quality data. This structured data in Excel, managed through VBA automation, allows for efficient organization, and facilitates downstream applications. The visualization step is developed using Power BI, which is directly connected to this central Excel database to provide dynamic dashboards for real-time monitoring of quality performance. This interactive dashboard presents key insights, including supplier performance, defect trends, inspection times, and weekly quality summaries, enabling quick decision-making.

To keep the systems applicable and user-oriented, a monitoring and feedback loop is integrated into the architecture. Thus, end-users, like inspectors, quality engineers, and production managers, could provide any feedback on system usability, data accuracy, and visualization clarity. This feedback is incorporated into further refining the ETL process, improving report design, as well as altering the system to the changing needs of inspection. Further, role-based access control is enforced in the system to guarantee data security and integrity, allowing only authorized personnel to access certain functionalities or data views. The architecture is designed for scalability, enabling future integration with other data sources, such as IoT sensors or real-time machine logs.

3.5.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed and understand the pattern and trend in the IQC data. This phase provided insights that informed the structure and design of dashboard component to understand the overall quality performance to highlight areas required improvement. The analysis highlighted key supplier performance issues, including high rejection rates from certain vendors and Lot Acceptance Rate (LAR) over different month, which is inconsistencies in supplier reliability. Additionally, repeated Notice of Defect (NOD) cases for specific part types pointed or recurring quality problem that may require corrective action. These insights played a crucial role in shaping the structure of quality dashboards and in defining key performance indicators (KPIs) that support data-driven decision-making and continuous improvement in supplier quality management.

The heatmap visualization in Figure 4.1 represents the correlation matrix among various variables in the dataset and is used to identify the strength and relationship between them. In the chart, dark blue shade indicates stronger positive correlations, while lighter shade represents weaker or no correlation. From the the visualization, we can see that mrno, pono and remark are strongly correlated with one another, it suggests that these variables are related. Additionally, there is noticeable positive correlation between lot_size and v_sample_size. meanwhile, variables such as supp_code, supp_name, and remark show relatively weak correlations with both lot_size and v_sample_size, indicating limited direct relationships. This heatmap plays a critical role in exploratory data analysis by revealing variable relationships, helping identify potential redundancies, guiding feature selection for further analysis.

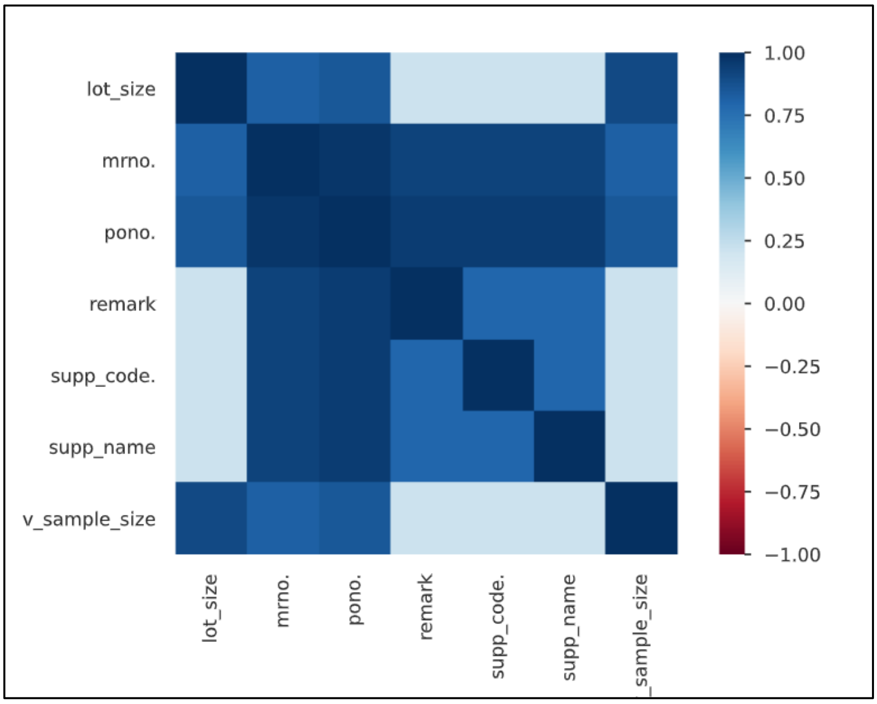


Figure 3. 5 Heatmap of Correlations

Figure 4.2 show that visual of supplier performance based on rejection data from the Incoming Quality Control (IQC) process. It highlights the ten-supplier performance with the highest total quantity of rejected item, it offer a quick insight into which vendor may be contributing most to quality issue. Each red bar represents a supplier, with height indicating the level of rejection. TT Electronics Integrated Manufacturing and Keymate Mechatronics (Suzhou) Co. Ltd stand out as the top two contributors to rejections, with significantly higher quantities than the rest, that potential ongoing quality problems. The remaining supplier have relatively lower and more consistent rejection quantities, suggesting that a few suppliers are primarily responsible for most issues.

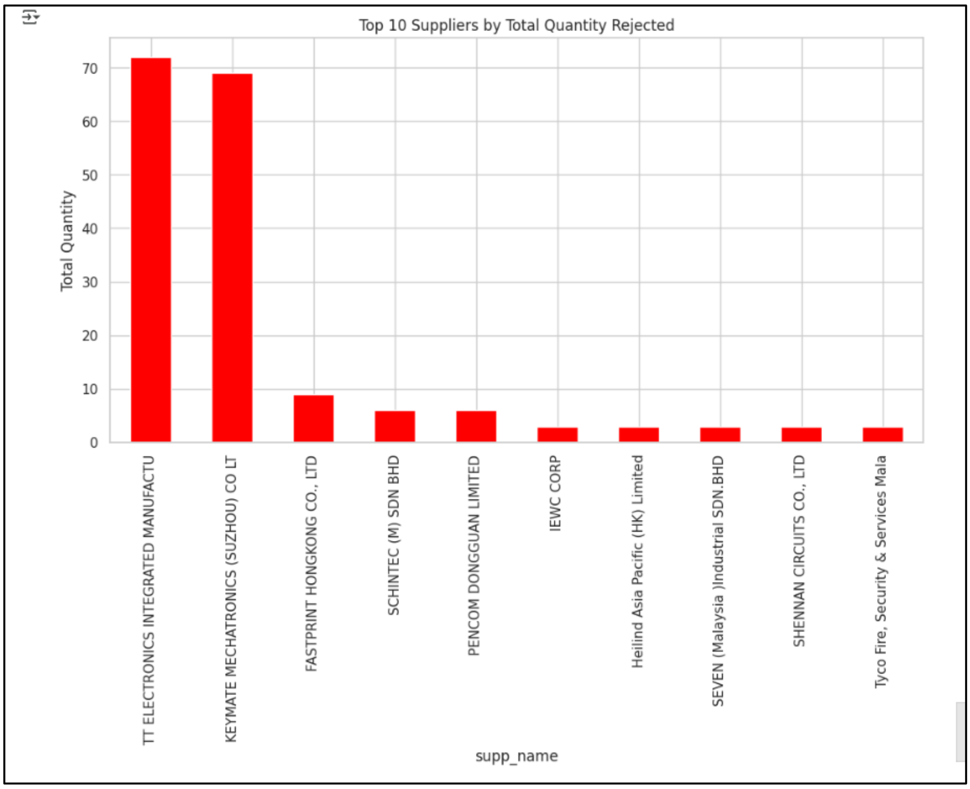


Figure 3. 6 Bar Chart of Top 10 Supplier by Total Quantity Rejected

This analysis supports the broader findings from the exploratory data analysis (EDA), which aimed to identify patterns and trends in supplier reliability and product quality. The visual insight gained from this chart plays a crucial role in shaping the quality dashboard design, informing key performance indicators (KPIs), and guiding strategic actions such as supplier audits, corrective measures, and continuous improvement initiatives in supplier quality management.

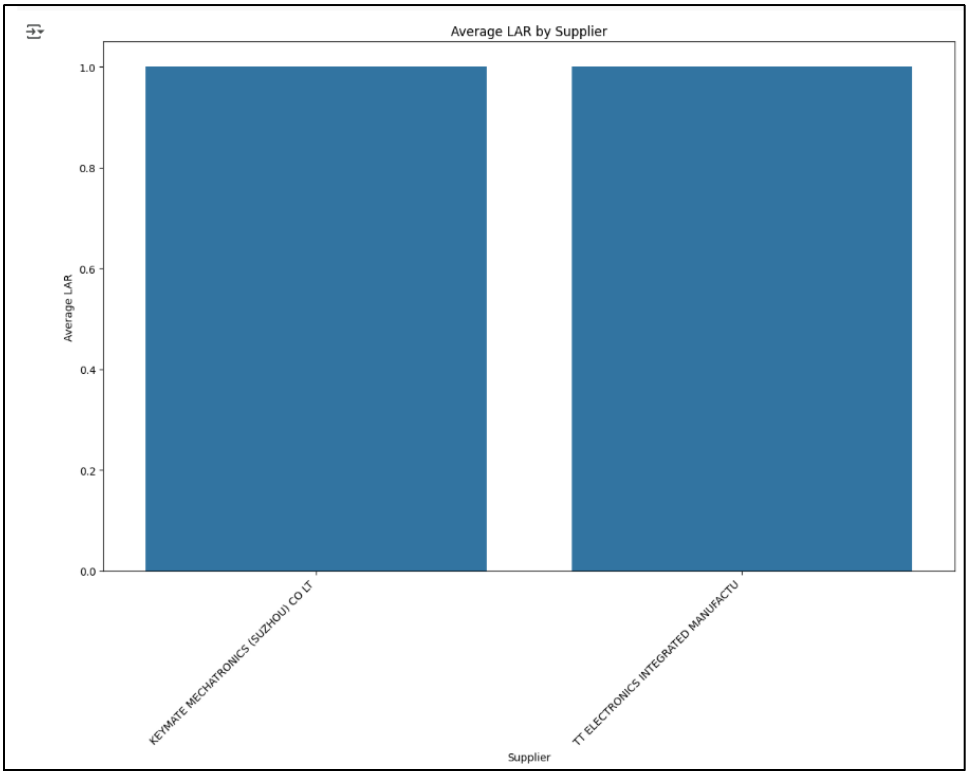


Figure 3. 7 Bar chart of Average LAR by Supplier

This bar graph, "Average LAR by Supplier," compares the average LAR for two different suppliers. LAR is probably an acronym for a performance indicator like Lead Time Adherence Rate or a comparable quality/delivery score. The suppliers are identified on the X-axis as "KEYMATE MECHATRONICS (SUZHOU) CO LT" and "TT ELECTRONICS INTEGRATED MANUFACTU," and the Average LAR is measured on the Y-axis on a scale of 0.0 to 1.0. The fact that both bars remarkably reach the greatest position on the Y-axis suggests that TT Electronics Integrated Manufacturing has attained an average LAR of roughly 1.0. This implies that both providers are operating at a very high and comparable level, effectively optimizing their performance in relation to the LAR measurement, according to this indicator. These insights formed the foundation for the development of automated quality reports and dashboard visualizations, as they guided the creation of key performance indicators (KPIs) used in the final system.

3.5.5 Predictive Modelling

In this study, predictive modelling is incorporated to enhance quality monitoring by anticipating potential defects based on historical inspection data. The modelling phase focuses on building machine learning models that can learn patterns from existing data to predict future inspection outcomes. Given the structure of the dataset comprising both quantitative and qualitative attributes such as defect type, lot size, inspection date, and supplier information classification models are suitable for predicting whether a lot will pass or fail inspection. The process begins by splitting the dataset into training and testing subsets, typically using an 80:20 ratio. Feature engineering is applied to convert categorical variables into numerical form through encoding techniques such as one-hot encoding or label encoding. Features are also normalized where necessary to ensure uniform scaling, which improves model performance.

The Random Forest Regressor demonstrated superior performance due to its ability to handle both linear and non-linear relationships, reduce overfitting through ensemble learning, and manage many input variables effectively. Hyperparameter tuning using grid search was applied to optimize model parameters such as the number of trees and maximum depth. This approach ensures that the final model not only fits the data well but also generalizes effectively to unseen data, making it reliable for predicting inspection results in real-time.

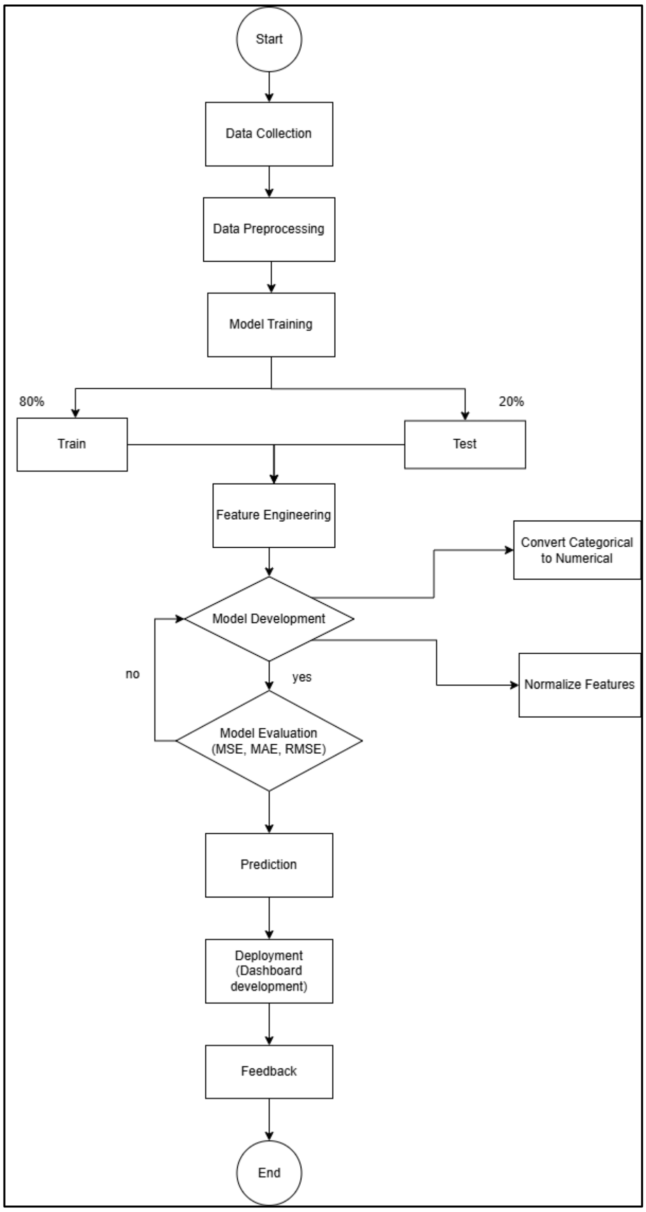


Figure 3.1 Flowchart of the Random Forest model

The prediction from the Random Forest model is calculated as the average output from all individual decision trees in the ensemble. Where $\hat{y}$ is the predicted value (in this case, the Predicted_Average_PPM), T is the total number of trees used in the model (n_estimators = 100), and $h_t(x)$ is the prediction made by the decision tree for the input x. The final prediction is obtained by averaging the results from all 100 trees, which helps reduce overfitting and improve overall prediction accuracy.

Prediction from Random Forest:

$$\hat{y} = \frac{1}{T}\sum_{t=1}^{T} h_{t(x)} \qquad (3.3)$$

Where:

$\hat{y}$: predicted value (Predicted_Average_PPM)

T: number of trees (n_estimator=100 in your model)

$h_{t(x)}$: prediction from the t-th decision tree for input x

Forecast for future:

$$\hat{y}future, i = \frac{1}{T}\sum_{t=1}^{T} h_{t(xfuture,i)} \qquad for\ i = 1, \ldots, 40 \qquad (3.4)$$

3.6 Improve Phase

In the Improve phase, the focus is on evaluating and selecting the suitable machine learning (ML) models to predict quality performance indicators such as defect rates and PPM (Parts Per Million).

3.6.1 Model Evaluation

To assess the performance of the trained Random Forest Regressor in predicting the Average of PPM (Parts Per Million) values, three standard regression evaluation metrics were computed: Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). These metrics quantify the difference between the predicted values and the actual observed values from the test dataset.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \qquad (3.5)$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i|)^2 \qquad (3.6)$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i|)^2} \qquad (3.7)$$

Where:

y_i Actual value for the i-th data point (Actual PPM)

$\hat{y}_i$: Predicted value for the i-th data point (Predicted Average PPM)

n: Total number of data points in the dataset

The MAE represents the average of the absolute value of the difference between the actual and predicted PPM values from the model. In simple terms, it gives an average idea of how much the model's prediction put the other way around from the real PPM value. Because MAE treats all errors equally and does not take direction into account, it is useful to gauge the typical size of errors in PPM units.

On the other, MSE computes the average of the squared differences between actual and predicted values. Given that the errors are squared, MSE stresses more on bigger difference measures as compared to MAE. Therefore, it's more sensitive toward outliers than the mean absolute error. When the model yields some large errors and goes a very long way in exceeding the bounds of expectations, the MSE will be greatly changed, thus specifying whether or not such extreme measures in prediction error should be taken as a concern.

Finally, and more simply, RMSE is the square root of MSE. Unlike MSE, it maintains the higher weightage on large errors but brings the result back in terms of measurement units, i.e. PPM, which makes RMSE more intuitive compared to MSE. RMSE, therefore, has a default use in regression problems and captures the different dimensions of the variances in prediction errors and their absolute magnitudes.

The Random Forest model achieved high accuracy and balanced performance across all metrics, making it the preferred model for deployment. Additionally, cross-validation was applied to assess the model's stability and reduce the likelihood of overfitting. The evaluation confirmed that the model is well-suited for integration into the automated quality monitoring system. Once deployed, it enables real-time predictions of inspection outcomes, allowing for early identification of high-risk lots and enabling timely corrective actions. Continuous feedback from actual inspection outcomes will be used to retrain the model periodically, ensuring its ongoing relevance and performance improvement over time.

3.7 Control Phase

In the Control phase, steps are implemented to guarantee that the enhancements are sustained over time. This involves developing standard operating procedures (SOPs), establishing automated update mechanisms, and consistently reviewing dashboard results. Ongoing feedback and evaluation systems assist in maintaining quality enhancements and promoting stability in processes over time.

3.7.1 Deployment

Deployment refers to the act of taking the finalized model and using it in production environments for decision-making or automated processes. In TT Electronics, this can involve setting the automated quality system on the quality control for production floor, where its primary function will be to continuously screen the quality of incoming materials and components, and issue real-time alerts to IQC operators regarding potential defects. By leverage AI tools to provide actionable insights, such as identifying defect trends it enables automated quality monitoring.

The system is developed specifically for quality control, ensuring that defect is detected in real-time. Additionally, it will generate statistical control chart to analyze production trends, visualize and identify variations that indicate potential quality issues.

The integration of the data pipeline into production environment is to ensure a seamless flow of data for real time analysis. The quality detection system is important component to enhance responsiveness in addressing potential quality issue. It used to notify operators about potential defect identified by the system. The user-friendly interface is developed to maximize usability. The goals is to enable user to easy interpreted data, monitor production quality and detect defect with the system effectively.

3.7.2 Feedback

The feedback is essential for the continuous improvement and long-term improvement of the system. It involves both technical performance evaluation and user experience assessment. This to monitoring the accuracy and reliability of the quality detection system in identifying defects and ensuring it meets the defined production goals, such as reducing rework, improving efficiency, and maintaining product quality. Feedback from key stakeholders, including operators, quality engineers, and production managers, who interact with the system daily. Their feedback provides valuable insights into system usability such as the user-friendly interface, the relevant visualized data and the ease of integration the system into their workflow. Using this feedback, iterative improvements are made to different system components. The data pipeline might be modified to incorporate new data sources or adapt to changes in data flow.

Table 3. 2 Shown question in feedback form

No	List of Questions
1.	How effectively does the system help you identify potential defects in incoming materials at an early stage?
2.	Is the visualization of quality data clear and helpful for decision-making?
3.	Does the system's real-time alerting mechanism improve your response time to quality issues?

3.8 Tools and library

This study use of variety data science and visualization tools and libraries. The main programming language is Python, and it use libraries such as Pandas and NumPy for data processing, Scikit-learn for machine learning applications and TensorFlow for complex modelling. Data visualization is done using python libraries like Matplotlib and Seaborn. In order to effectively manage and query big datasets and enable the development of the automatic quality detection system, Excel database that facilitated by VBA (Visual Basic for Applications).

3.9 Summary

This chapter has outlined the research methodology used to develop an automated quality detection system for TT Electronic. The research design covers the collection, preprocessing, and integration of production data into a centralized data warehouse using a robust ETL pipeline, ensuring efficiency and scalability. Finally, the deployment phase ensures seamless integration into TT Electronics' production environment, while the feedback mechanism enables continuous monitoring and improvement. Next, will continue to chapter 4 about expected outcome and conclusion.

CHAPTER 4

DATA ANALYSIS, RESULTS AND DISCUSSION

4.1 Introduction

This chapter will discuss the comprehensive data analysis conducted based on the objectives of this study. It consists of two main parts which are the data analysis and the discussions of results. The data analysis part explains the methodologies used to achieve the objectives, with the details of processes, challenges and techniques applied. The chapter is divided into sections that reflect each phase of the process automation, reporting, and visualization. Finally, discussions are conducted to explain how the objectives are achieved throughout the study.

4.2 Exploratory Data Analysis

This material trend a bar and line graph, shows the monthly variations in the "Count of Material" received and the "Count of Completed Insp Date," which represent inspection activity. There is a high link between January and April, with the brown bars signifying material count and the blue line denoting completed inspections precisely aligning. After reaching a peak of 1861 units in February, activity gradually decreased in March and April. May saw a significant comeback, with the material count rising to 1492 and almost all of the materials (1486) being inspected. This steep drop in June, in contrast to the other months, calls for more research since it might be a sign of a seasonal slowdown, an interruption in the material supply chain, or an inadequate data set for the month. The fact that 181 materials fell under the first "(Blank)" category indicates that some products were handled without a designated month but nonetheless underwent examination. All things considered, the graph offers a clear visual depiction of material flow and inspection efficiency, showing a steady processing rate for most of the time, with a notable outlier in June.

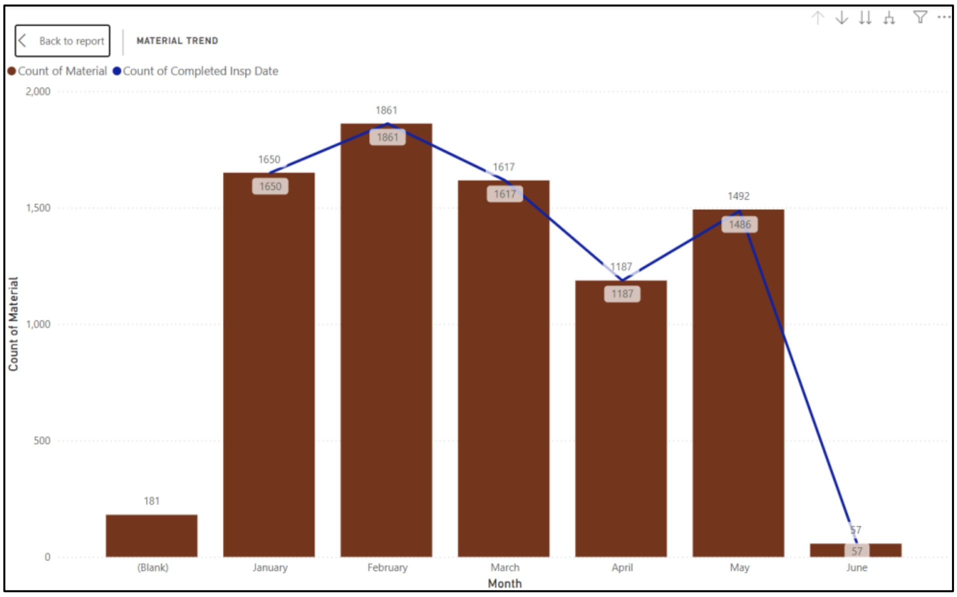


Figure 4. 8 bar and line graph of material trend

This line graph, pending material clearly shows a worrying pattern in the number of items that are pending examination. The Y-axis, "Count of Material," measures the backlog, while the X-axis, "Inspection Month," shows a chronology from February to May 2025. The Inspection Status: Pending, shown by the blue line, shows a notable and ongoing rise in uninspected items throughout these months. The pending material count increased marginally to about 28 in March 2025 from a comparatively low count of about 23–24 in February. But in April 2025, there was a notable spike that raised the number to almost 41. The pending material count peaked on this graph in May 2025, near 95, because of this sharp increase.

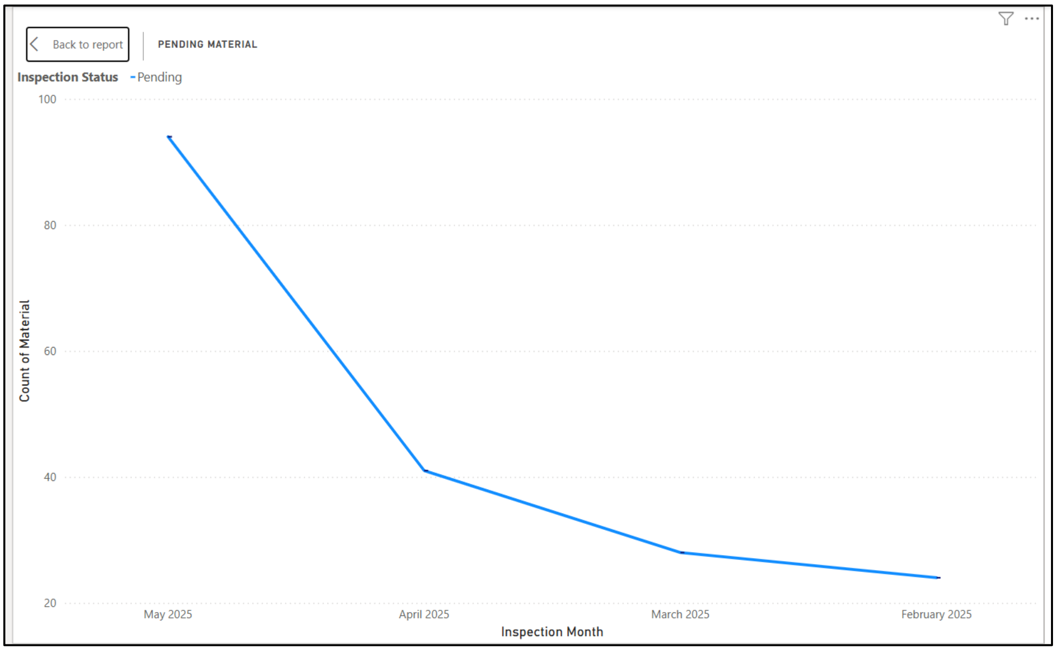


Figure 4. 9 Line chart of pending material

4.3 ETL Pipeline Development

To the automation and real-time reporting objectives 1 of this study, a custom ETL (Extract, Transform, Load) pipeline was developed. The ETL pipeline serves as the backbone of the IQC quality data monitoring system, ensuring that inspection data flows seamlessly from raw Excel files to a centralized reporting structure used in Power BI dashboards.

The extraction phase focused on retrieving data from multiple Excel files, each representing different product categories and inspection periods. These files were automatically detected and loaded from predefined directories using Python's and glob libraries. The pandas.read_excel() function I Figure 4.4 was used to load relevant sheets into Data Frames which is database.

```
data = pd.read_excel("IQC_Q1.xlsm", sheet_name='Database')
data.head()
```

Figure 4. 4 Load data using Python

The transformation stage was the most critical component of the pipeline. Data from various sources was cleaned, standardized, and enriched to ensure consistency and usability for analysis and reporting as shown on Figure 4.5.

```
#drop no relevent column
data = data.drop(['Number','InvNo.', 'M Sample Size', 'Finding', 'MRB No.', 'NOD No.', 'SCAR No.'], axis=1)
data.head()
```

Figure 4. 5 Drop not relevant column

This line is part of data cleaning, it removes columns from the dataset that are not relevant to your analysis or dashboard reporting. This helps reduce noise and improves processing efficiency. The line of code shown in Figure 4.6 is used to standardize and clean the column names in the dataset to ensure consistency throughout the analysis. It applies three transformations to every column name in the DataFrame. It converts all characters in the column names to lowercase using lower(), ensuring uniformity and reducing the chances of case-sensitive errors in later stages of processing. This step is particularly important when working in Python.

```
[11] # Ensure consistency in column names and formats
     data.columns = [col.strip().replace(' ', '_').lower() for col in data.columns]
```

Figure 4. 6 Standardize column name

In the final phase of the ETL (Extract, Transform, Load) pipeline, the processed and cleaned data was exported into various formats to support automated reporting and dashboard integration. This "Load" step ensures that the refined dataset is readily accessible for both real-time analytics and stakeholder communication.

One of the primary targets for data export was CSV (Comma-Separated Values) files, which are widely compatible and serve as an efficient medium for integrating with Power BI. Power BI can directly connect to these CSV files to refresh and display up-to-date inspection results and quality metrics in a visual, interactive dashboard.

Furthermore, the data pipeline was designed with scalability, including an optional SQL database structure. While not yet fully implemented, this setup prepares the system for future integration with a relational database, enabling real-time querying, centralized data management, and advanced analytics through direct SQL queries. This forward-thinking design supports long-term sustainability and enhances the potential for live data monitoring and decision-making in IQC operations.

4.4 Data Transformation and Centralization

This data transformation and centralization is the techniques employed to transform unprocessed inspection data from various origins into an organized, and interpretable format. It specifically explains how Microsoft Excel and VBA (Visual Basic for Applications) macros were utilized to create an automated ETL (Extract, Transform, Load) pipeline. Integrating data from inspection checklists and reports into a centralized IQC database streamlines the process by removing manual data entry, decreasing error risks, and maintaining consistent formatting for subsequent analysis and reporting, has effectively achieved Objective 2. The subsequent subsections detail the technical execution, automation procedure, user interface, and change management strategy that facilitated the successful implementation and deployment of the automated IQC data management system.

4.4.1 VBA Macros in Excel

Visual Basic for Applications (VBA) is a programming language built into Microsoft Excel that allows users to automate repetitive tasks, enhance functionality, and create customized solutions through "macros." A macro is a set of programmed instructions that can execute tasks such as data entry, formatting, calculations, and file operations. VBA enables users to write custom scripts that interact with Excel objects like worksheets, ranges, cells, and buttons. This makes it a powerful tool for building dynamic solutions such as automated reports, data extractors, and interactive dashboards.

While Microsoft Excel is not a relational database system like Microsoft Access or SQL-based platforms, it can be structured to function as a basic database, particularly for small to medium-sized datasets. Each row can represent a unique record, while columns serve as fields or variables. With proper data organization, validation, and automation, Excel can be used effectively for data collection, storage, and analysis. However, it is essential to maintain data consistency, use unique identifiers (like MR number or P.O. number), and avoid manual errors, which can be achieved through automation.

4.4.2 Automation of IQC Database Using VBA

In the context of this study, VBA macros were employed to automate the process of extracting, transforming, and loading (ETL) inspection data into a centralized Excel-based IQC database. This automation eliminates the need for manual copying and pasting of data from various files, significantly reducing the risk of human error and saving time.

The system uses two key source files the MR file and the IQC Checklist. The VBA macro extracts required variables from these files, standardizes their format, calculates quality metrics such as PPM (Parts Per Million) and LAR (Lot Acceptance Rate), and prepares them for insertion into the main database file, 'IQC Report Database - EMS Year 2025.xlsx'.

To simplify user interaction, an "Insert" button is provided within the Excel dashboard as shown in Figure 4.7. When the user clicks this button and selects the inspection date, the VBA script is triggered. The script automatically reads relevant data from the source files, matches and merges the values based on predefined logic, and locates the next available empty row in the database sheet as shown in Figure 4.8. The new data is then inserted vertically, creating a new record in the database without disturbing existing records.

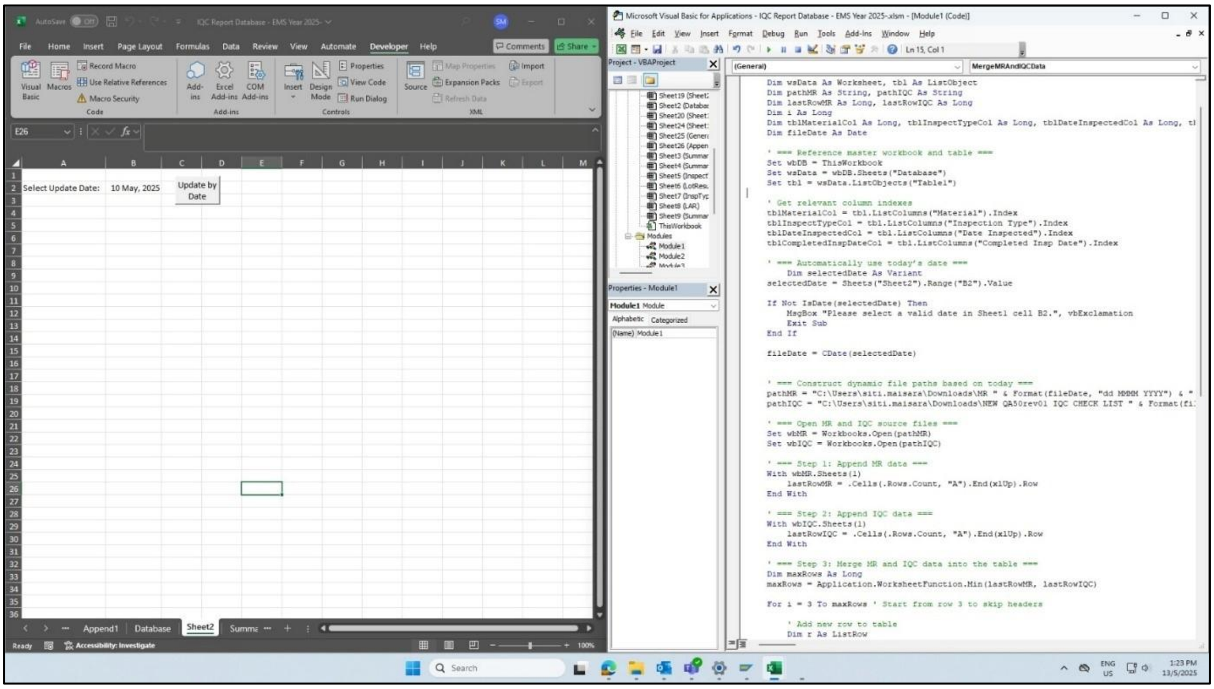


Figure 4. 10 "Insert" button is provided within the Excel dashboard and VBA code

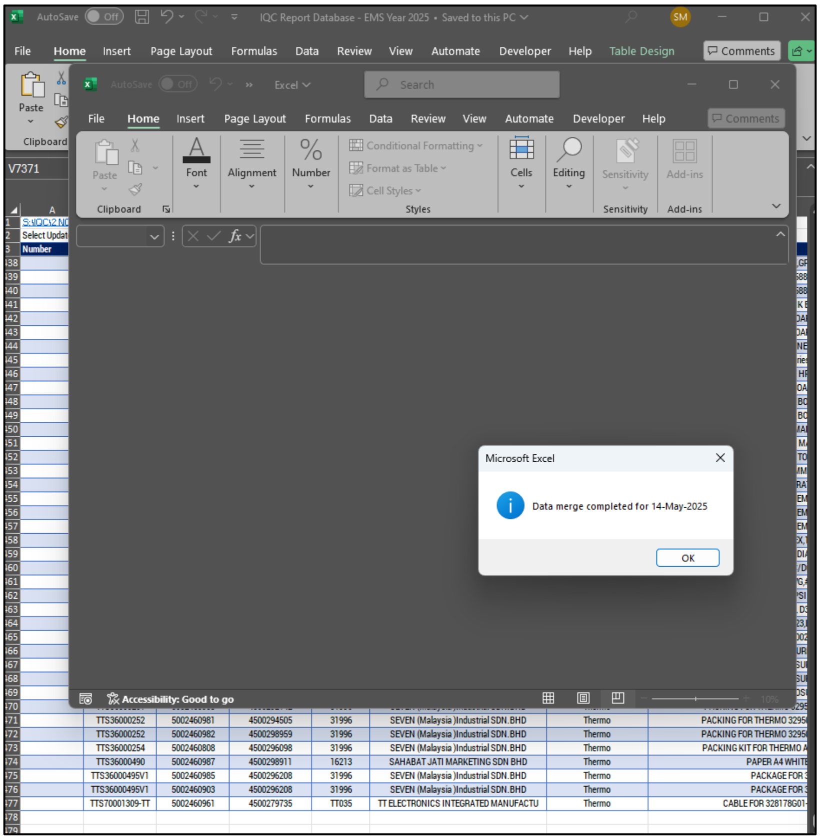
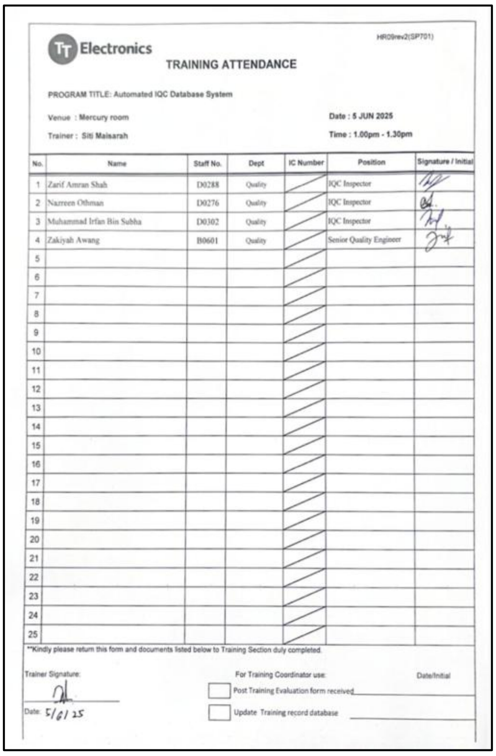


Figure 4. 8 Two file being merge.

This automated solution ensures real-time updating of the IQC database, supports consistent formatting, and enables accurate tracking of quality performance across multiple categories. The result is a user-friendly, efficient, and scalable data management system tailored for manufacturing quality control operations as shown in Figure 4.9.

Figure 4. 11 The Automated Database

Beyond technical development, the successful implementation of this automated IQC database also involved PDCA stands for Plan-Do-Check-Act, a cyclical quality improvement method. After developing and validating the macro, we conducted a briefing session with inspectors and quality engineers to demonstrate how to use the system. During the sessions, explained how the source files had to be structured, the steps to run the macro using the Insert button and emphasized the importance of consistent data formatting. I offered detailed instructions on how to use the system, including how to name the source files, what sheets had to be named, which fields were mandatory, and how to use cell references that had been standardized to allow the automation to run without any issues. Following the PDCA cycle, the training and briefing sessions have been completed to implement and standardize the process. This standardization is essential to maintain data integrity and ensure the automated system operates smoothly. Figure 4.10 provides evidence that this database has been successfully implemented within the company.



TT Electronics

HR09rev2(SP701)

TRAINING ATTENDANCE

PROGRAM TITLE: Automated IQC Database System

Venue : Mercury room

Date : 5 JUN 2025

Trainer : Siti Maisarah

Time : 1.00pm - 1.30pm

No.	Name	Staff No.	Dept	IC Number	Position	Signature / Initial
1	Zarif Amran Shah	D0288	Quality		IQC Inspector	
2	Nazreen Othman	D0276	Quality		IQC Inspector	
3	Muhammad Irfan Bin Subha	D0302	Quality		IQC Inspector	
4	Zakiyah Awang	B0601	Quality		Senior Quality Engineer	
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**Kindly please return this form and documents listed below to Training Section duly completed.

Trainer Signature:

Date: 5/6/25

For Training Coordinator use: Date/Initial

☐ Post Training Evaluation form received

☐ Update Training record database

Figure 4. 120 Training Attendance for Automated IQC Database

After the deployment of the automated IQC database system in IQC department, continuous feedback will be sought from the users, including inspectors and quality engineers, to measure the solution's effectiveness, usability, and reliability. Whenever any situation, such as a deviation, an inconsistency, or an area needing improvement, is observed from day-to-day use, immediate corrective action will be taken. Regular monitoring and periodical review shall be followed to maintain its capability in delivering quality data management in an accurate, efficient, and standardized manner towards the continuous improvement of TT Electronics.

Since the implementation of this system, the data collection process for IQC has been faster, more reliable, and easier to manage. Inspectors and quality personnel are thus free to devote time to quality analysis rather than data entry. The automated system works for outputs structured data that go into real-time Power BI dashboards to monitor quality performances.

4.5 Predictive Modelling for Defect Part Per Million (PPM)

To address the challenge of predicting whether incoming materials should be accepted or rejected during the IQC process, a classification model was developed. The predictive modelling aimed to assist in automating the decision-making process, reduce inspection errors, and enhance quality assurance consistency be archived for objective 3.

The Random Forest Regressor is a machine learning model that combines the predictions of multiple decision trees, making it robust to noise and overfitting while capturing complex nonlinear relationships. It is initialized with specific hyperparameters (n_estimators=100, max_depth=10) to balance complexity and generalization. For multi-step future forecasting, the model uses a sequential prediction approach. When actual lag data is unavailable like future dates, previously predicted PPM values are used to generate lagged features, allowing for iterative prediction over an extended time horizon. The model was implemented using Python and the scikit-learn library as shown in Figure 4.11.

```
[32] # Initialize Random Forest
     model = RandomForestRegressor(
         n_estimators=100,
         max_depth=10,
         min_samples_split=5,
         min_samples_leaf=2,
         random_state=42,
         n_jobs=-1
     )

     # Train model
     print(" Training Random Forest model...")
     model.fit(X_train, y_train)

     # Make predictions
     train_pred = model.predict(X_train)
     test_pred = model.predict(X_test)

     print(f"\nModel Train Score: {model.score(X_train, y_train):.4f}")
     print(f"Model Test Score : {model.score(X_test, y_test):.4f}")

      Training Random Forest model...

     Model Train Score: 0.5796
     Model Test Score : -21.4316
```

Figure 4.11 Develop Random Forest Model

The input dataset contained in a DataFrame called data was a pre-processing to reveal the proper temporal alignment. The values within the column 'MR_Date' signified the time of accepting a material. They had to be converted to datetime and set as an index for the DataFrame so that time-series-aware modelling methods would apply while respecting the events' chronological sequence.

The target variable to be predicted is "Average_of_PPM", the historical defect rate per million units. All other inputs include all the available numerical and categorical columns except for the actual target that were to be included as model inputs. These features, in turn, may represent variations in process, supplier, and material characteristics that may affect inspection outcomes.

To enhance validation of robustness, 5-fold cross-validation is employed on the training data with cross_val_score. This splits the data into five segments, training the model on four while validating on the fifth in succession, aiding in maintaining performance uniformity across various data subsets. Moreover, feature importance analysis is performed utilizing the Random Forest’s inherent functionality, providing insights into which have the greatest impact on PPM predictions. or testing. A 100-tree Random Forest Regressor was therefore fit on the historical data. This model was chosen because of its resistance to overfitting and its capacity to find complex, nonlinear relationships without any need for scaling features. Once trained to predict the PPM values on the testing dataset, common regression metrics such as RMSE or MAE can be used to evaluate the model's performance and accuracy of prediction.

4.5.1 Model Evaluation

To assess the performance of the Random Forest Regressor model in predicting the Average of Parts Per Million (Average_of_PPM), three standard regression evaluation metrics were used: Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) as shown in Figure 4.12. The assessment of the Random Forest Regressor model provides significant insights regarding its effectiveness on both the training and testing datasets. The Mean Absolute Error (MAE) for the training set is 11.1093, suggesting that on average, the model's predictions deviate by approximately 11 units from the true PPM values. Yet, the MAE rises considerably to 20.5579 on the test set, indicating that the model is less precise with data it hasn't encountered before. Likewise, the Root Mean Squared Error (RMSE) is 18.2797 for the training set and rises to 22.0601 for the test set. Because RMSE imposes greater penalties on larger errors than MAE, the rise in RMSE suggests that the model is making notably larger errors on the test dataset. The Mean Squared Error (MSE), being the square of the RMSE, also demonstrates this trend showing values of 334.1458 (train) and 486.6487 (test). The variations in performance between training and testing clearly show that the model is overfitting it effectively learns the training data but has difficulty generalizing to new, unseen data.

Evaluation Metrics:

	Metric	Train	Test
0	MAE	11.1093	20.5579
1	RMSE	18.2797	22.0601
2	MSE	334.1458	486.6487

Figure 4. 12 Model evaluation results

Despite this, the model still demonstrates a reliable level of accuracy in forecasting PPM values. To evaluate its forecasting capability, the model was applied to predict the monthly average PPM values for the year 2025. These monthly predictions as shown in Figure 4.13 reflect a relatively consistent quality performance trend throughout the year, with values ranging between 46.90 to 55.11 PPM. This stability suggests that no significant fluctuations in defect rates are expected under current conditions.

PREDICTED MONTHLY AVERAGES (2025):

	Month	Average Predicted PPM
0	Jan	46.90
1	Feb	52.80
2	Mar	54.63
3	Apr	55.11
4	May	53.98
5	Jun	53.61
6	Jul	54.27
7	Aug	53.43
8	Sep	54.13
9	Oct	53.82
10	Nov	52.79
11	Dec	54.09

Figure 4. 13 PPM Prediction results

Overall, the model demonstrates a reliable level of accuracy for forecasting PPM values. The relatively small gap between MAE and RMSE suggests that while the model is generally accurate, there may be a few instances of larger errors. These findings confirm that the Random Forest model is suitable for short-term forecasting of quality performance and can support proactive decision-making in Incoming Quality Control (IQC).

4.6 Dashboard Visualization

To meet the fourth objective, a real-time and interactive dashboard was developed using Power BI. The dashboard provided a user-friendly interface for visualizing key quality metrics from the IQC inspection data.

The IQC Dashboard in Figure 4.14 provides a clear and comprehensive view of the quality inspection process and performance. The Incoming Quality Control (IQC) Dashboard presents a detailed perspective on inspection performance at TT Electronics, emphasizing key trends in supplier quality and the disposition of incoming materials. Current data indicates that a total of 8,045 materials have been received however, upon inspection, only about 7,858 materials were inspected, showing a very high rate of completion of inspections.

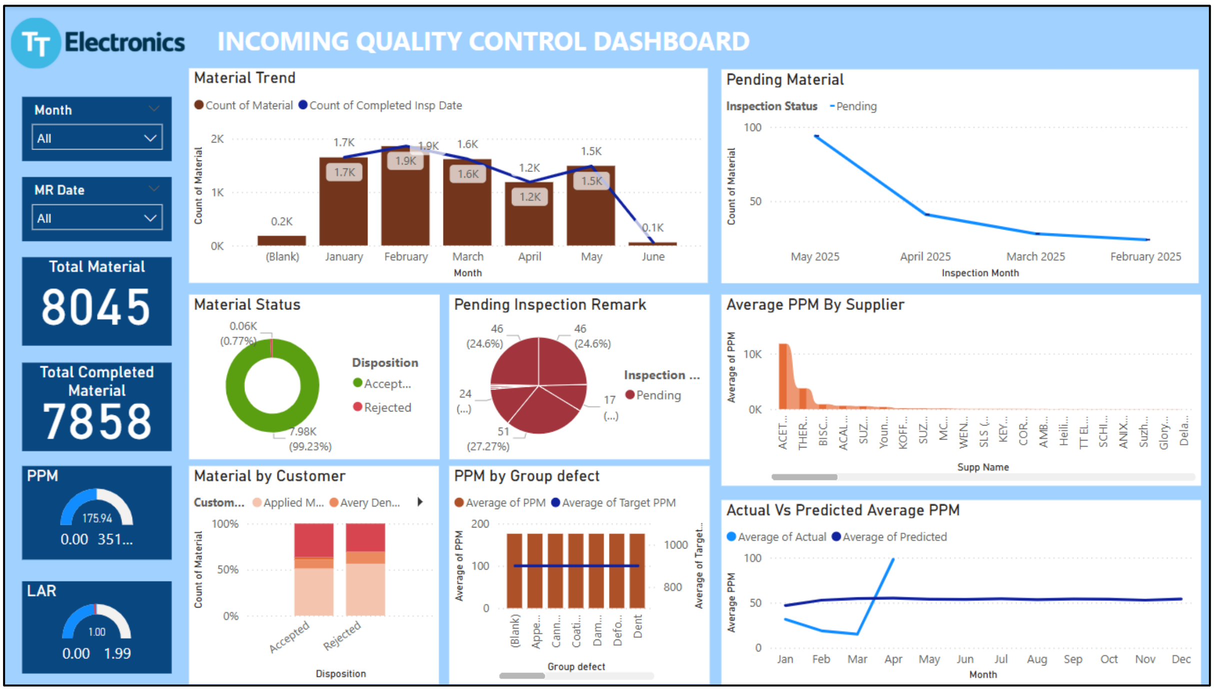


Figure 4. 14 IQC Dashboard

The Material Trend view shows the maximum number of material receipts occurred in February (1.9K materials) and hence a continuous decrease toward June, though the steep decline in June may indicate either less incoming material or some incomplete data entry. The trend further indicates that the materials received are also strongly aligned with materials inspected upon receipt, thereby ensuring timely inspection processing.

The Average PPM by Supplier chart shows which suppliers are sending more defective parts. For example, suppliers like ALCETRO and BISCO. have higher PPM values, which may require closer monitoring or follow-up with those suppliers. With the Pending Material chart, the backlog for inspections is seen diminishing with time, with pending counts having dropped from May going backward to February, thus implying increased efficiency in the inspection process. Material disposition-wise, as given in the Material Status pie chart, 99.23% of all the materials inspected were accepted and only 0.77% were rejected, indicating the consistently high quality from suppliers. However, the Pending Inspection Remark chart states that there still remain a small number of items (17) pending, with others requiring follow-up or clarification-an area worthy of consideration.

Another useful chart is Count of Material by Customer, which shows how much material each customer is receiving and how much was accepted or rejected. This helps in tracking customer-specific quality. The Material Status donut chart shows that overall, 99.26% of materials are accepted and only 0.74% are rejected, which reflects strong quality control performance. Lastly, the Actual vs Predicted PPM chart shows that actual PPM values are lower than or close to the predicted values, meaning the system is performing well and forecasting is reliable.

Overall, this dashboard helps identify what is working well such as low rejection rates and fewer pending inspections and also highlights areas that need attention, like defect types and certain suppliers. It allows the quality team to make quick decisions based on real-time data and supports continuous improvement in the inspection process.

The line chart in Figure illustrates the comparison between actual and predicted values of Average of PPM using the Random Forest Regressor model. The solid blue line represents the actual PPM values observed during the test period, while the dashed orange line shows the model's predictions.

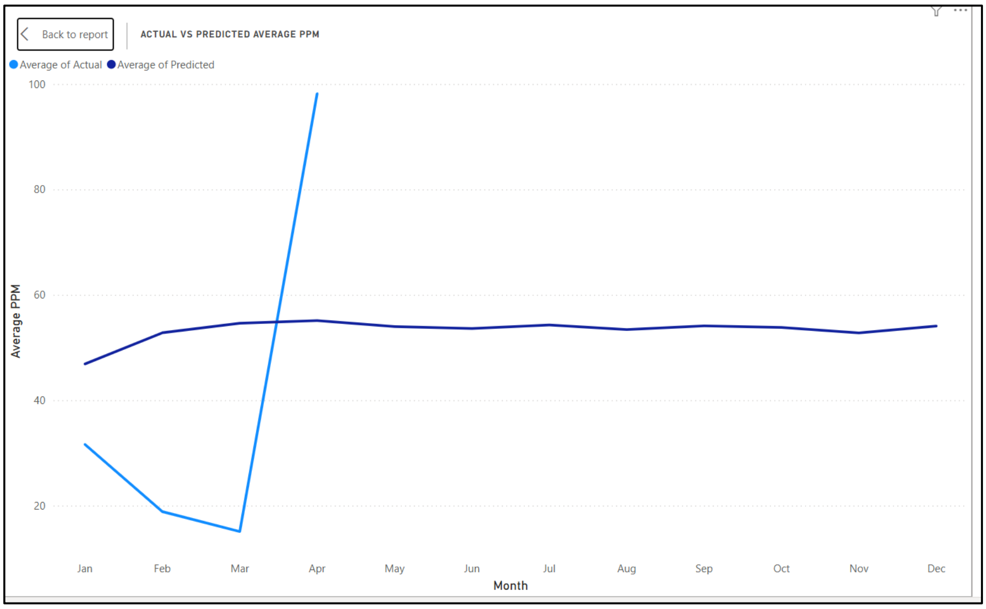


Figure 4. 15 Line chart of Actual Vs Predicted Average PPM results

The chart shows the comparison between the actual PPM (Parts Per Million) and the predicted PPM over a period of months. We can observe that in April, the actual PPM shows a sharp increase. This happened because the quantity of materials received in April was lower compared to previous months, but the number of rejected items was relatively high.

The predicted PPM line tends to be more stable against changes from month to month. This stability arises because the predictive model, say Random Forest or another type of regression-based model, considers historical trends and selected features while forecasting. These models are usually trained to minimize the overall error and thus focus more on generalizing the pattern over time, making them less able to react to a one-time drop in material quantity or an unexplained rise in rejections. This shows the difference, that actual PPM is based on operational parameters such as quantity of materials being received, and real-time rejections, while predicted PPM represents expected defect trends given past data behavior. Hence big-off-the-mark or outlier events like the sharp rise observed in April might simply never be captured by the modelling unless such circumstances were part of the historical training dataset. Therefore, while the model can be rightly used to identify general trends for planning, it may not always fully represent rare or extreme cases.

Therefore, the analysis concludes that the Random Forest model is good for prediction quality trends for a month, as these trends have been reversed by average PPM in the past and hence may be relied upon to forecast near-future outcomes towards which decision-making can proceed favouring the IQC process.

4.7 Summary

Chapter 4 has been elaborating on the result of developing and implementing the Quality Data Monitoring System for Incoming Quality Control (IQC) at TT Electronics. It began with data collection and preparation from various IQC sources in a way that it was clean and correct. Through data analysis, meaningful trends in supplier performance and defects were determined that helped guide the system design. A key deliverable was building an automated ETL pipeline. The tool leverages VBA and Python to transfer data from Excel files to a central database, which is connected to Power BI for reporting. This has made IQC data handling more efficient, accurate, and less manual.

Apart from that, a Random Forest Regressor model was built to predict the Average Parts Per Million (PPM) so that trends in the quality of incoming material can be anticipated. The model performed well and assists in decision-making at an early stage. Finally, an interactive Power BI dashboard was created to display IQC results in real time. To conclude, the chapter illustrates that all the research objectives were achieved, and the system introduced provides a useful and automated way of monitoring and improving incoming quality control.

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

5.1 Introduction

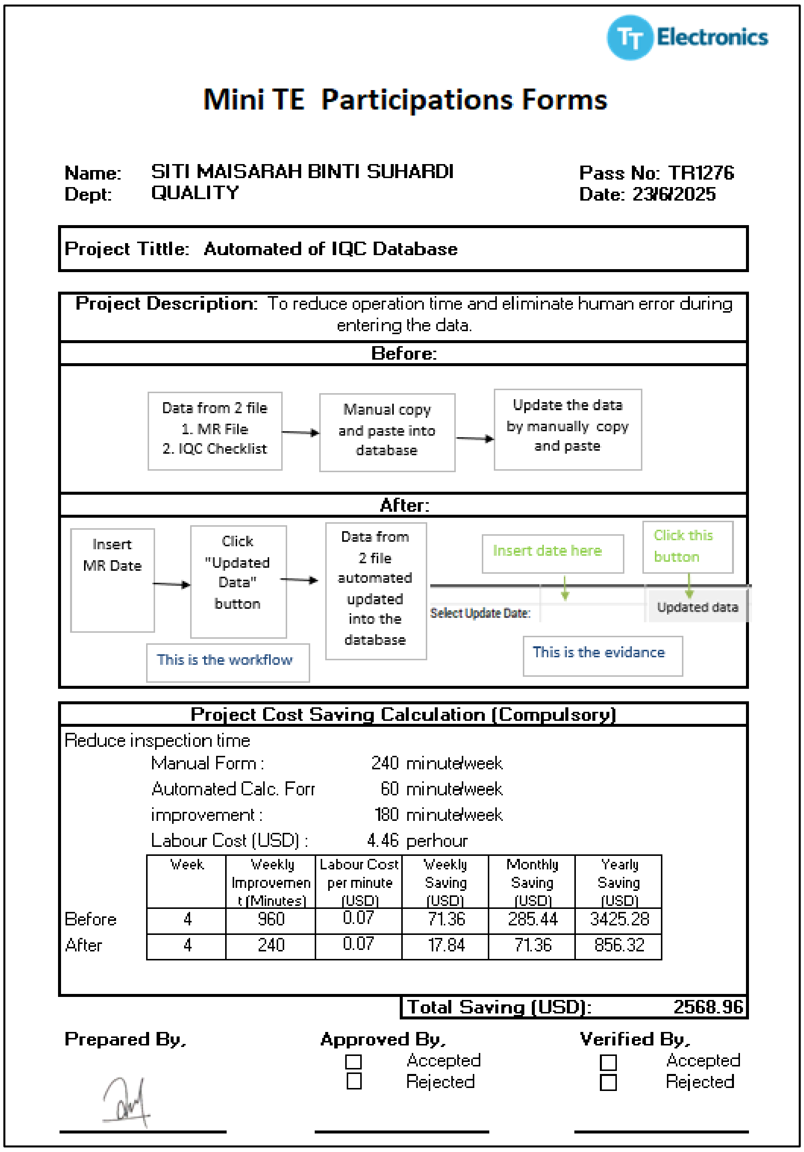
This concluding chapter summarizes the entire research is drawing together the key findings and achievements of the "Implementation of a Quality Data Monitoring System for Incoming Quality Control (IQC) in Manufacturing" at TT Electronics. Building upon the detailed analysis and results presented in Chapter 4, this chapter will reiterate how the research objectives were met and underscore the significance of the developed system. It will also address the limitations encountered during the study and briefly explore an entrepreneurship idea stemming from this project. Furthermore, a set of actionable recommendations for future work will be provided, outlining potential enhancements and expansions to ensure the long-term sustainability and maximize the impact of the quality data monitoring system. The chapter aims to offer a holistic perspective on the project's contributions and its implications for continuous quality improvement within TT Electronics' manufacturing operations.

5.2 Summary of study

The study developed and implemented a Quality Data Monitoring System for Incoming Quality Control (IQC) at TT Electronics, altering the manual quality control procedures into a fully automated, data-based one. The study followed the y integrating the Data Science Life Cycle and the DMAIC (Define, Measure, Analyze, Improve, Control) framework. in which it was realized that there was an urgent need for enhanced quality assurance in manufacturing, given the long manual processes prone to errors.

The system automation of IQC inspections was done by means of Excel VBA macros, thus fulfilling the first objective. These macros had a very disciplined approach in capturing inspection data and greatly reduced time and the human error that accompanied the manual filling of inspection forms. The automation standardized the input process, thus enabling a precise and rapid recording of IQC data from different inspection sources with minimum human intervention. Before automation, the manual process via forms would consume around 240 minutes a week for just data handling. After automation, this was reduced to just 60 minutes per week, achieving a weekly time improvement of 180 minutes. With a labor cost of USD 4.46 per hour (USD 0.07 per minute), this reduction translates into a weekly cost saving of USD 71.36, and an annual saving of approximately USD 3,425.28. This system is well worth investing in now and into the future because it meets one of the key requirements for long-term quality assurance-checking while reducing inspection time and labor costs.

The goal to centralize IQC data was accomplished through the development and implementation of a custom ETL pipeline. This pipeline extracted data from MR files and IQC checklists, converted them into a standardized format, and loaded them into the central IQC database. Developed using Python libraries of Pandas and NumPy and complemented with VBA for a smooth Excel automation flow, the process guaranteed data integrity



Mini TE Participations Forms

Name: SITI MAISARAH BINTI SUHARDI — Pass No: TR1276
Dept: QUALITY — Date: 23/6/2025

Project Tittle: Automated of IQC Database

Project Description: To reduce operation time and eliminate human error during entering the data.

Before:

After:

Project Cost Saving Calculation (Compulsory)

Reduce inspection time

Manual Form :	240	minute/week
Automated Calc. Forr	60	minute/week
improvement :	180	minute/week
Labour Cost (USD) :	4.46	perhour

	Week	Weekly Improvement (Minutes)	Labour Cost per minute (USD)	Weekly Saving (USD)	Monthly Saving (USD)	Yearly Saving (USD)
Before	4	960	0.07	71.36	285.44	3425.28
After	4	240	0.07	17.84	71.36	856.32

Total Saving (USD): 2568.96

Prepared By, — Approved By, ☐ Accepted ☐ Rejected — Verified By, ☐ Accepted ☐ Rejected

Figure 5.1 Evidence project cost saving calculation

Thirdly, employing a Random Forest Regressor fulfilled the goal of enhancing predictive capabilities. The model was trained on historical IQC data with the goal of forecasting the average PPM values monthly. Evaluation metrics such as MAE, RMSE, and MSE depicted a capable prediction model. The forecast for 2025 showed a consistent monthly forecast of base PPM values, which could facilitate making decisions ahead of time, thus enabling the IQC team to forecast and tackle potential quality defects in advance.

Fourthly, an interactive Power BI dashboard is developed to enable stakeholders for real-time monitoring and visualization of IQC data. Stakeholders could track inspection results, defect rates, supplier performances, and predicted PPM values atop these from this dashboard, all in a single centralized view. This enhanced visibility could lead to quicker response to quality deviations and better resource planning.

5.3 Limitation of the study

Various limitations were revealed during the study, despite the successful implementation of the Quality Data Monitoring System. The first limitation is that the Excel-based database system, albeit VBA macro-enhanced, limits scalability and does not facilitate querying as efficiently as an actual relational database like SQL Server or PostgreSQL.

Secondly, third-party inspectors are still manually entering some IQC data, which may allow human error and inconsistencies to creep in. Such manual entry required the team to spend a lot of time cleaning and preprocessing the data. Thirdly, the predictive modelling remained quite limited in scope. The Random Forest Regressor considered primarily the structured numerical data while disregarding unstructured textual data such as defect remarks. Then, advanced forecasting techniques or external influencing factors were not considered.

And finally, though the system does support real-time reporting from the Power BI side, it is yet to push the next level by linking to real-time data streams from IoT sensors or manufacturing machines. This, along with that, would push further accuracy and timeliness in monitoring.

5.4 Entrepreneurship Idea

This study has the potential to be expanded as a business opportunity. Businesses may greatly increase operational efficiency and cut expenses by incorporating predictive analytics and machine learning will further enhance the system's capabilities by predicting potential defects before they occur. Companies can cut waste, minimize production downtime and optimize resource allocation, which will result in considerable cost saving and improve operational efficiency. Real-time data collection and monitoring on the manufacturing floor will be possible through integration with IoT-enabled devices. Real-time monitoring reduces delays caused by undetected defects and ensures that quality issues are addressed instantly.I oT integration also makes it possible for businesses to collect vast amounts of high-quality data, which can enhance machine learning models for predictive analytics.

5.5 Recommendations

To help with sustainability and ongoing improvements to the Quality Data Monitoring System, several recommendations are made. First, the recommendation is made to establish a common template for the automated IQC database so that it can be used in future years. This template must contain instructions on the file names, data entry formats, and standard variable structures. This would provide a uniform environment in which future users can update and improve the system without major changes. It would also assist in slashing user training time while helping the data integrity and continuity in the long run.

On the predictive side, recommendation is to set up time-series forecast models that would integrate into the system. While the Random Forest model, on the other hand, provides good short-term forecasts of PPM, time-series models such as Holt-Winters, ARIMA, LSTM, or GRU would do a better job of tracking trends and seasonal movements of defect rates in time. Such an implementation would increase the capability of the system to foresee quality shifts of a longer duration and facilitate data-based planning and preventative action.

5.6 Conclusion

In conclusion, this study has successfully demonstrated the feasibility and significant benefits of implementing a Quality Data Monitoring System for Incoming Quality Control (IQC) at TT Electronics. By transitioning from manual processes to an automated, data-driven approach, the project has established a robust framework that encompasses efficient data management via an ETL pipeline, insightful data analysis through EDA, accurate defect prediction via machine learning, and comprehensive visualization through interactive dashboards. This system not only enhances the efficiency and accuracy of IQC operations but also provides the necessary tools for proactive quality management, empowering TT Electronics to make informed decisions and continuously improve the quality of its incoming materials. The successful completion of this project lays a strong foundation for future advancements in smart manufacturing and sets a precedent for leveraging advanced analytics in industrial quality assurance. While current limitations highlight areas for further development, the fundamental capabilities established position TT Electronics for a competitive advantage in maintaining high-quality standards.

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APPENDICES

Appendix A: Data Analysis and preprocessing

https://colab.research.google.com/drive/117hNuOPhEFnRDVisbhEjcwztnwy3oAC?usp

Appendix B: Modelling and Evaluation

https://colab.research.google.com/drive/model_rf

Appendix C: VBA Code Automated IQC Database

```
Sub MergeMRAndIQCData()

    Dim wbDB As Workbook, wbMR As Workbook, wbIQC As Workbook
    Dim wsData As Worksheet, tbl As ListObject
    Dim pathMR As String, pathIQC As String
    Dim lastRowMR As Long, lastRowIQC As Long
    Dim fileDate As Date, selectedDate As Variant
    Dim monthFolder As String
    Dim i As Long

    ' === Performance Boost ===
    Application.ScreenUpdating = False
    Application.Calculation = xlCalculationManual
    Application.EnableEvents = False

    On Error GoTo CleanupError

    ' === Reference master workbook and table ===
    Set wbDB = ThisWorkbook
    Set wsData = wbDB.Sheets("Database")
```

```
    Set tbl = wsData.ListObjects("Table1")

    selectedDate = wsData.Range("B2").Value
    If Not IsDate(selectedDate) Then
        MsgBox "Please select a valid date in Sheet 'Database' cell B2.", vbExclamation
        GoTo CleanupExit
    End If

    fileDate = CDate(selectedDate)
    monthFolder = Format(fileDate, "MM MMMM")

    pathMR = "W:₩Senior Quality Engineer₩00 KPI₩01 Daily₩02 IQC₩IQC 2025₩DAILY IQC₩DAILY MR₩" & monthFolder & "₩MR " & UCase(Format(fileDate, "dd MMMM yyyy")) & ".xls"
    pathIQC = "W:₩Senior Quality Engineer₩00 KPI₩01 Daily₩02 IQC₩IQC 2025₩DAILY IQC₩DAILY IQC₩" & monthFolder & "₩" & _
              "NEW QA50rev01 IQC CHECK LIST " & CStr(Day(fileDate)) & "-" & UCase(Format(fileDate, "MMM")) & "-" & Year(fileDate) & " FORMULA_update newNEW.xlsx"

    If Dir(pathMR) = "" Then MsgBox "MR file not found: " & pathMR, vbCritical: GoTo CleanupExit
    If Dir(pathIQC) = "" Then MsgBox "IQC file not found: " & pathIQC, vbCritical: GoTo CleanupExit

    Set wbMR = Workbooks.Open(pathMR, ReadOnly:=True)
    Set wbIQC = Workbooks.Open(pathIQC, ReadOnly:=True)

    ' Get last data rows
    With wbMR.Sheets(1)
        lastRowMR = .Cells(.Rows.Count, "A").End(xlUp).Row
    End With
```

```
    With wbIQC.Sheets(1)
        lastRowIQC = .Cells(.Rows.Count, "A").End(xlUp).Row
    End With

    ' Calculate row counts
    Dim startRowMR As Long: startRowMR = 2
    Dim startRowIQC As Long: startRowIQC = 4

    Dim rowCountMR As Long: rowCountMR = lastRowMR - startRowMR + 1
    Dim rowCountIQC As Long: rowCountIQC = lastRowIQC - startRowIQC + 1
    Dim maxRows As Long: maxRows =
Application.WorksheetFunction.Min(rowCountMR, rowCountIQC)

    If maxRows < 1 Then
        MsgBox "No data found in source files.", vbExclamation
        GoTo CleanupExit
    End If

    ' === Load MR and IQC data into arrays ===
    Dim mrData As Variant, iqcData As Variant
    mrData = wbMR.Sheets(1).Range("A" & startRowMR & ":M" & (startRowMR +
maxRows - 1)).Value2
    iqcData = wbIQC.Sheets(1).Range("G" & startRowIQC & ":H" & (startRowIQC +
maxRows - 1)).Value

    ' === Prepare output array ===
    Dim numCols As Long: numCols = tbl.ListColumns.Count
    Dim outputData() As Variant
    ReDim outputData(1 To maxRows, 1 To numCols)

    For i = 1 To maxRows
```

```
        ' MR data
        outputData(i, tbl.ListColumns("Material").Index) = mrData(i, 1)
        outputData(i, tbl.ListColumns("Description").Index) = mrData(i, 2)
        outputData(i, tbl.ListColumns("MRNo.").Index) = mrData(i, 4)
        outputData(i, tbl.ListColumns("MR Date").Index) = mrData(i, 6)
        outputData(i, tbl.ListColumns("Lot Size").Index) = mrData(i, 7)
        outputData(i, tbl.ListColumns("PONo.").Index) = mrData(i, 11)
        outputData(i, tbl.ListColumns("Supp Code.").Index) = mrData(i, 13)

        ' IQC data
        outputData(i, tbl.ListColumns("Inspection Type").Index) = iqcData(i, 1)

        Dim inspectedDate As Variant
        inspectedDate = iqcData(i, 2)
        If IsDate(inspectedDate) Or IsNumeric(inspectedDate) Then
            outputData(i, tbl.ListColumns("Date Inspected").Index) = inspectedDate
            outputData(i, tbl.ListColumns("Completed Insp Date").Index) = inspectedDate
        End If
    Next i

    ' === Write all rows in one go ===
    Dim insertStart As Range
    If tbl.DataBodyRange Is Nothing Then
        Set insertStart = tbl.Range.Cells(2, 1)
    Else
        Set insertStart = tbl.DataBodyRange.Cells(1,
1).Offset(tbl.DataBodyRange.Rows.Count)
    End If
    insertStart.Resize(maxRows, numCols).Value = outputData
```

```
    MsgBox "Data merge completed for " & Format(fileDate, "dd-mmm-yyyy"),
vbInformation

CleanupExit:
    On Error Resume Next
    If Not wbMR Is Nothing Then wbMR.Close False
    If Not wbIQC Is Nothing Then wbIQC.Close False
    Application.ScreenUpdating = True
    Application.Calculation = xlCalculationAutomatic
    Application.EnableEvents = True
    Exit Sub

CleanupError:
    MsgBox "An error occurred: " & Err.Description, vbCritical
    Resume CleanupExit

End Sub
```